

Multimedia

Vorlesung

Hannes Kaufmann

Research Unit Virtual and Augmented Reality
Institute of Visual Computing & Human-Centered Technology

Based on material by Prof. Christian Breiteneder

Psychoacoustics

- Scientific study of sound perception and audiology: How humans perceive various sounds
- **Study on your own** the Youtube course “Psychoakustik” (German)
https://www.youtube.com/watch?v=8_TxY93L8xE&list=PLftcEygAp_6lk8SpKNesLim_omxfXMbmo&index=1

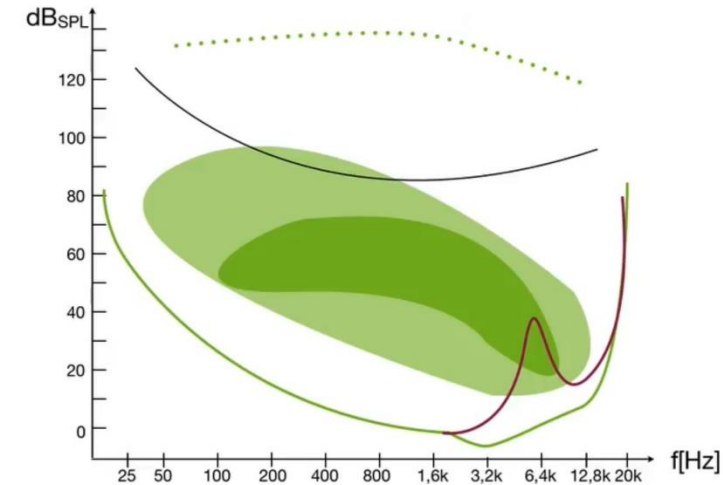
11 parts, ~ 107 min (total)

Inverted Classroom: Psychoacoustics Questions

- How do we hear?
- What happens inside the cochlea?
- What are the different ways to judge the location & direction of sound?
- What is 1 MEL? What is 1 BARK?
- What is critical bandwidth?
- How large is a critical band?
- What is the connection between critical bands and hair cells?

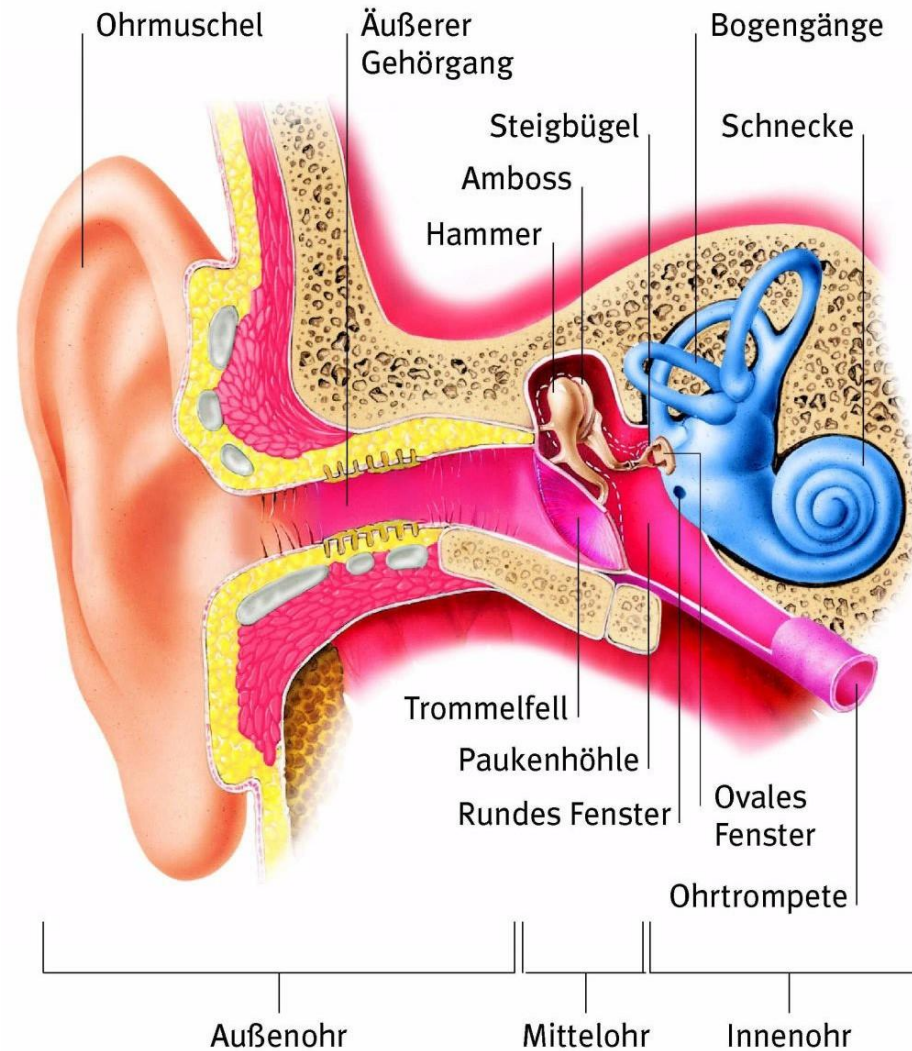
Inverted Classroom: Psychoacoustics Questions

- Explain and annotate the figure ->
- What is masking?
- What is temporal masking?
- Describe a psychoacoustic phenomenon (e.g. Shepard Scala, Tritonus Paradoxon,...)



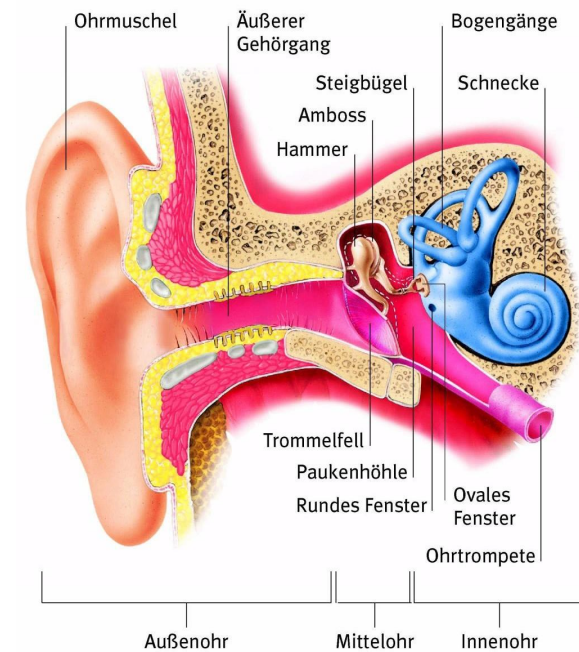
Psychoacoustics - Summary

The Human Ear

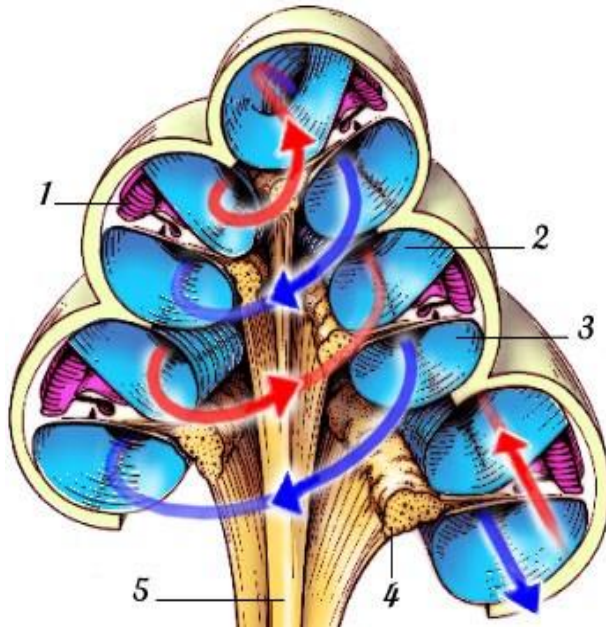


Hearing

- Sound waves hit the eardrum
- Eardrum transmits vibration to the ossicular chain in the middle ear;
 - as the last link the foot plate (“Steigbügel”) and the
 - oval window are moved.
- Thereby oscillation of the endolymph in scala vestibuli



Cochlea



1. cochlear duct
2. scala vestibuli
3. scala tympani
4. spiral ganglion
5. auditory nerve fibres

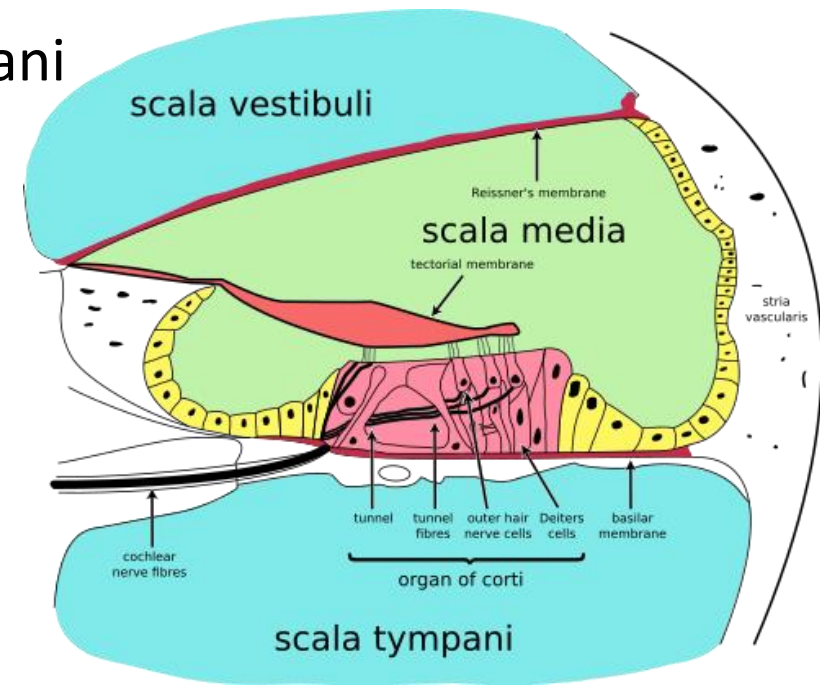
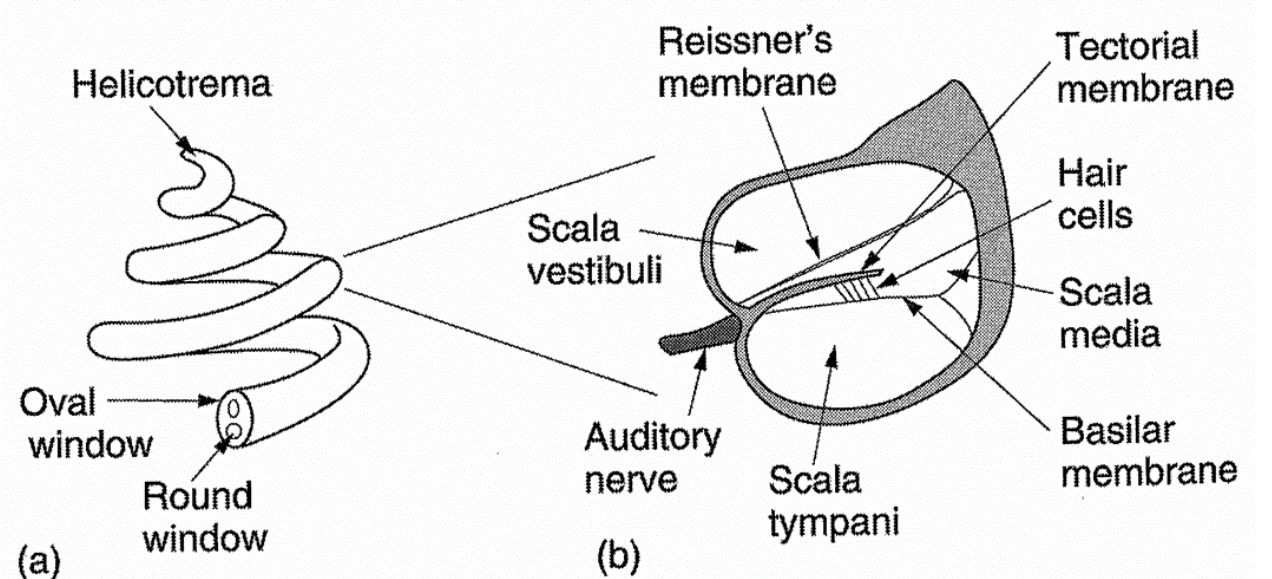


2 mm

- The red arrow is from the oval window
- The blue arrow points to the round window
- The cochlea is about 2 mm in diameter

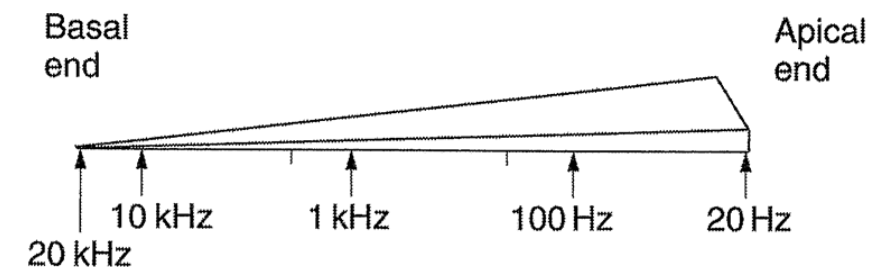
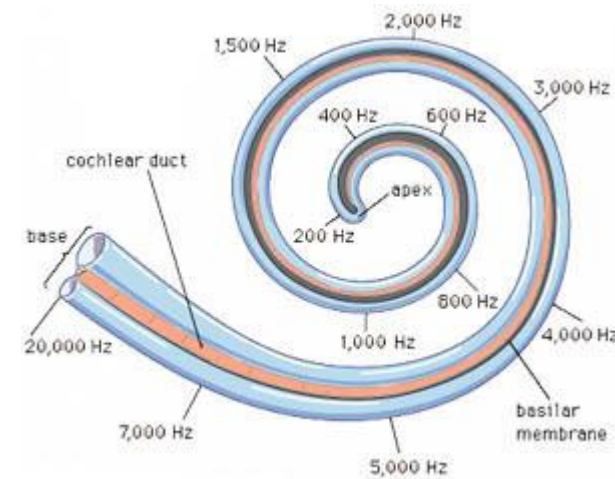
Hearing Cochlea

- 3 superimposed fluid-filled passages:
- *Scala vestibuli* (vestibule staircase)
- *Scala media* or cochlear duct (spiral duct) and
- *Scala tympani* (tympani staircase)
- Basilar membrane separates Scala Media and Scala tympani
- On basilar membrane approx. 15,000 hair cells
- *Outer hair cells* (3 rows) serve to amplify sound traveling waves within the cochlea; prefilter.
- *Inner hair cells* (1 row) perform conversion of mechanical oscillations into nerve impulses (transduction).

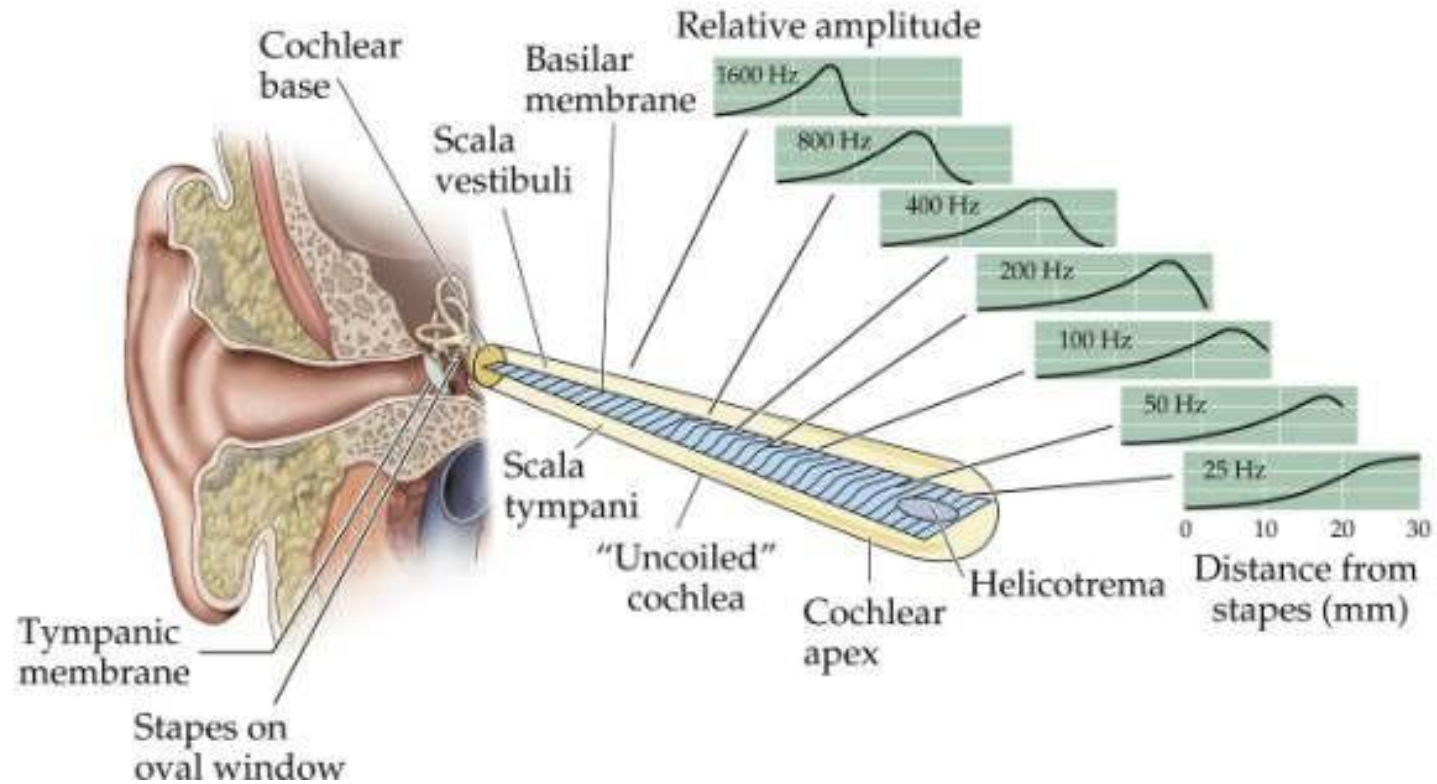


Basilar membrane

- Width of basilar membrane increases from oval window to helicotrema.
- This changes the mechanical properties (mass loading, stiffness, damping) and thus also the vibration properties of the system as a function of the distance to the helicotrema.
- Basilar membrane resonates at different points for different frequencies.
- High frequencies have a deflection maximum near the oval window, low frequencies, on the other hand, only near the helicotrema.

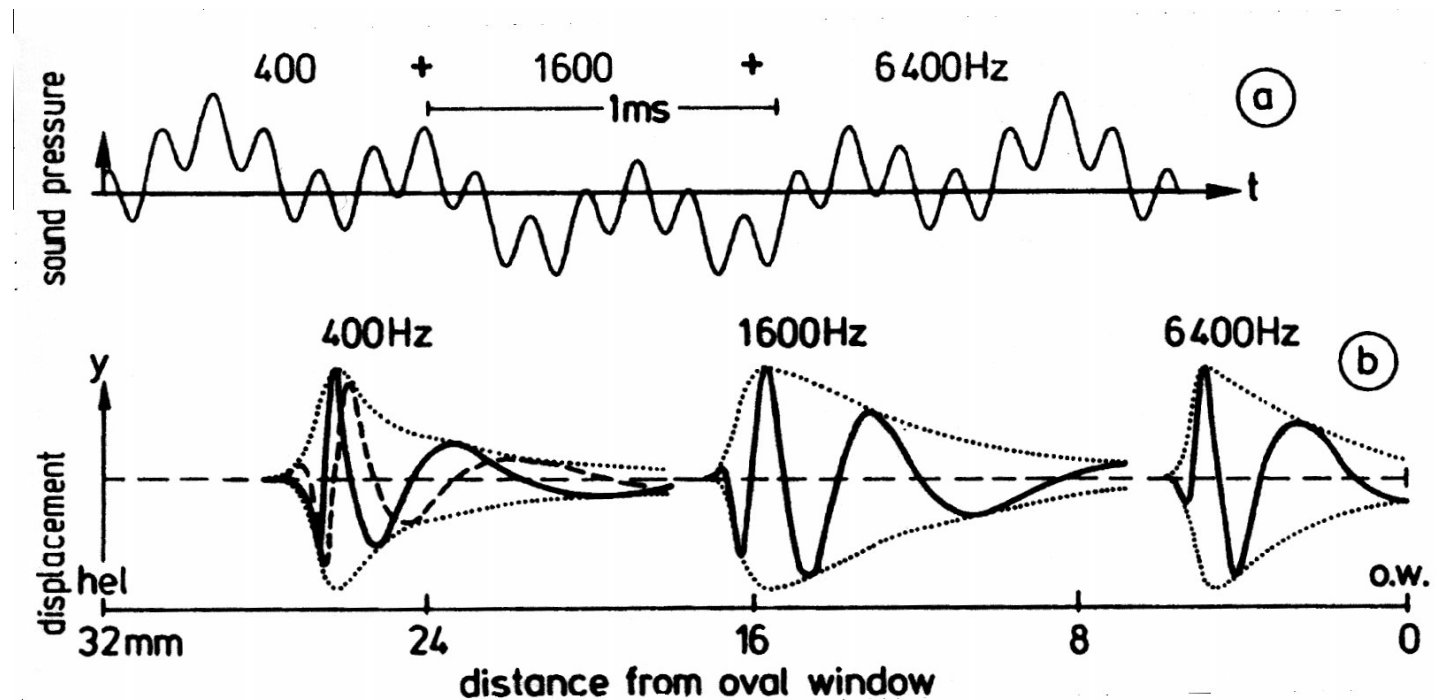


Basilar membrane

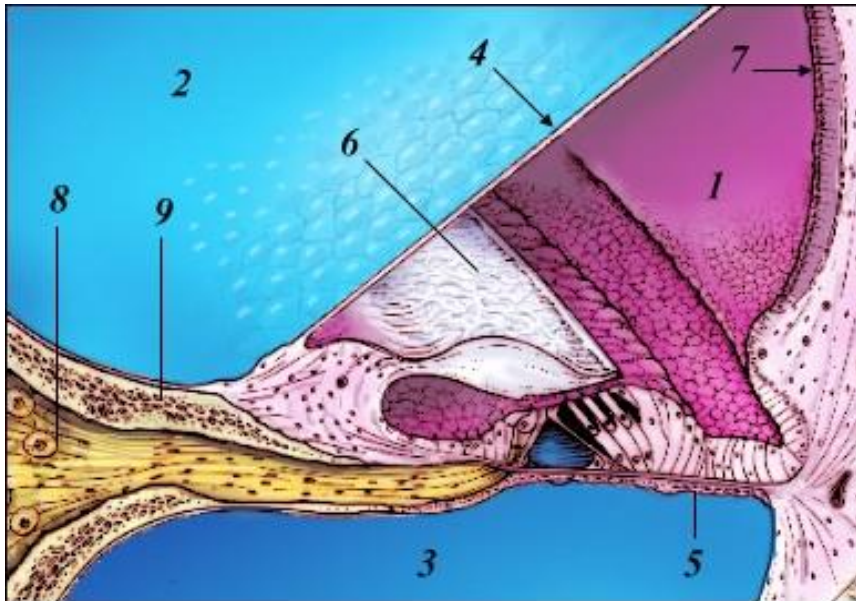


Basilar membrane

Frequency-to-place transformation:



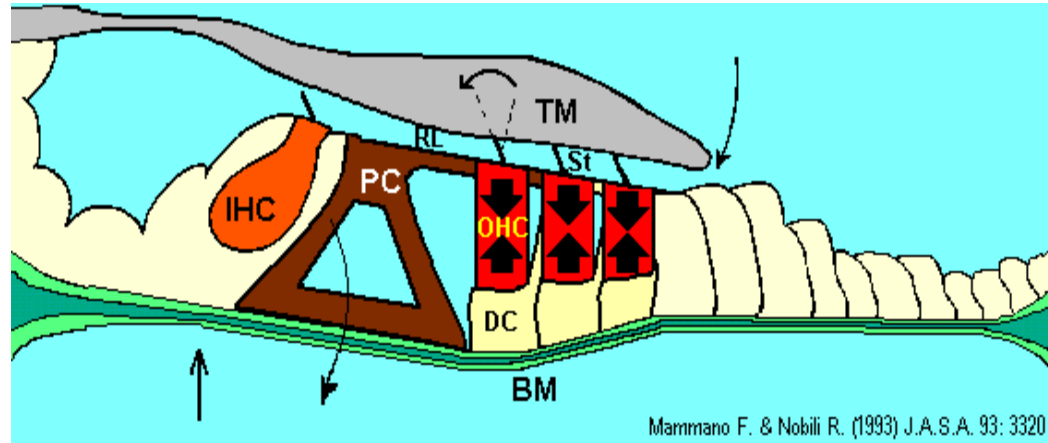
Cholea



1. Cochlear duct
2. Scala vestibuli
3. Scala tympani
4. Reissner's membrane
5. Basilar membrane
6. Tectorial membrane
7. Stria vascularis
8. Nerve fibres
9. Bony spiral lamina

The organ of Corti is on top of the basilar membrane
under the tectorial membrane

Organ of Corti



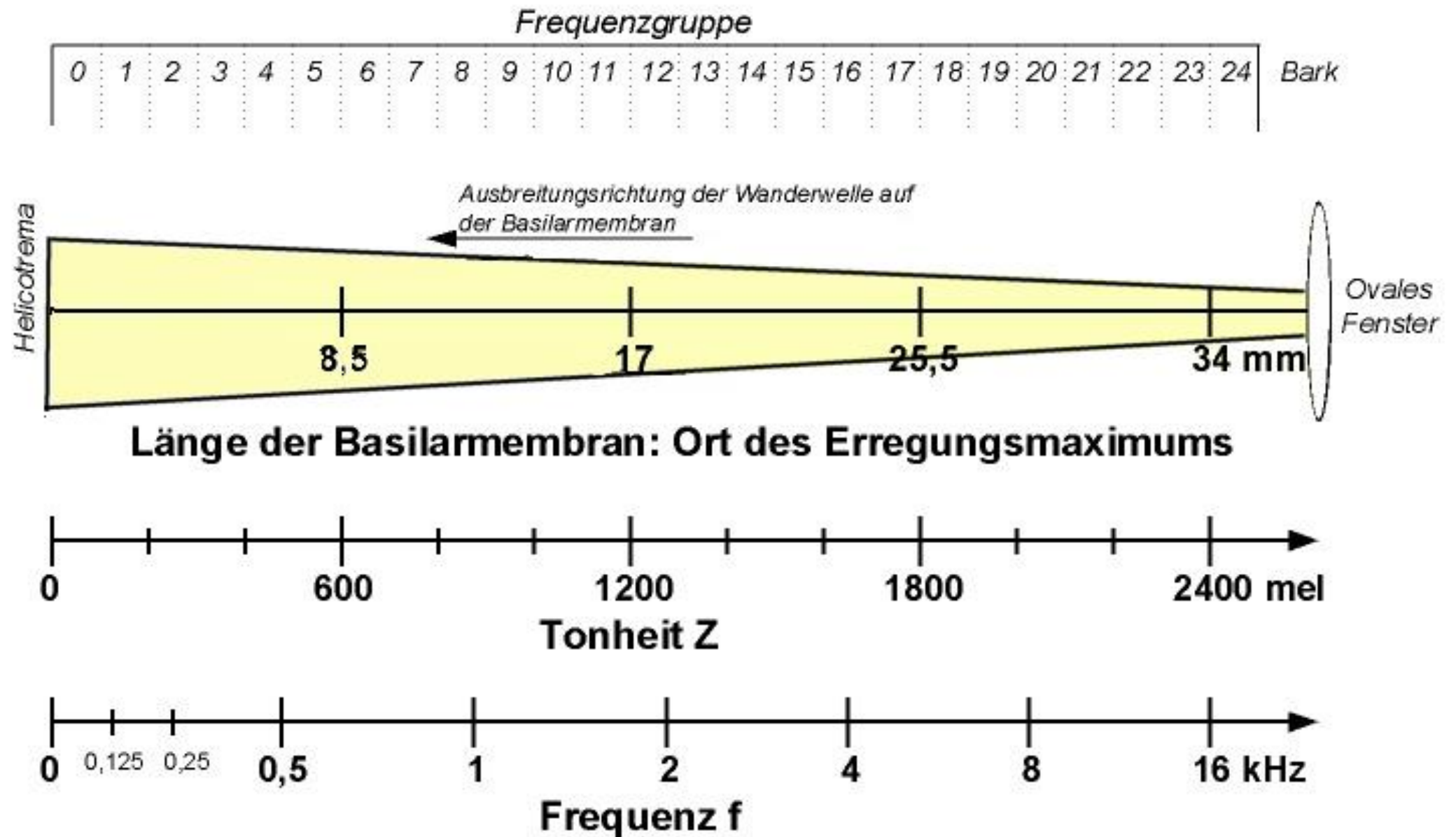
- Tectorial membrane (TM)
- Basilar membrane (BM)
- Inner haircells (IHC)
- Outer haircells (OHC)
- Deiters' cells (DC)
- Reticular lamina (RL)
- Pillar cells (PC)
- Stereocilia (St)

Hair Cells

- At the resonance point, the vibrations are amplified and thus the inner hair cells are stimulated more strongly.
- Beyond the resonance point, the oscillations are strongly damped, the corresponding frequency hardly propagates further (frequency selectivity)
- High frequencies (resonance point near the oval window) do not cause excitation of the inner hair cells for low frequencies,
- low frequencies, however, also excite the hair cells responsible for high frequencies.

Frequency Groups, Critical Bands

- Hearing divides the basilar membrane into **24 sections of equal length** = **frequency groups**.
- **Nerve impulses from a frequency group are evaluated together ([pitch], volume, sound, direction).**
- Width of a frequency group is about 100 Hz for frequencies up to 500 Hz and a ~100Hz – 4kHz above 500 Hz (minor third = “kleine Terz”). Corresponds to approx. 1 Bark = 100 Mel.
- **Critical Band / Critical Bandwidth** is not fixed (e.g. 100-200Hz) but each tone is creating a critical band around itself.
- Unlike critical bands, which are based on psychoacoustic principles, frequency groups are more of a practical construct used for organizing and processing audio signals.



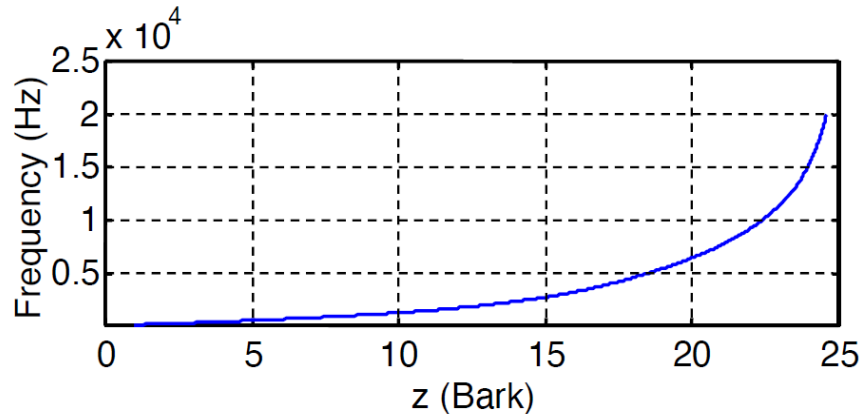
1 Bark = 100 Mel = 1,3mm on the basilar membrane = 150 hair cells = 27 JND
Verdopplung in Mel entspricht Verdopplung der wahrgenommenen Tonhöhe.

Bark Scale and Critical Bandwidth

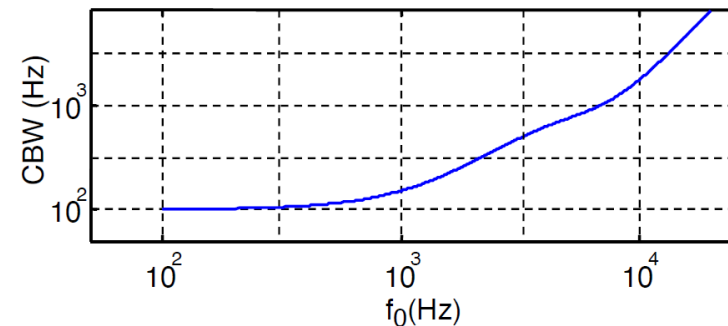
- A scale that converts frequency (Hz) into units of critical bandwidth

$$z(f) = 13 \arctan(0.00076 f) + 3.5 \arctan \left[\left(\frac{f}{7500} \right)^2 \right] \quad (\text{Bark})$$

named after Heinrich Barkhausen (who proposed the first subjective measurements of loudness)



Bark scale



Critical bandwidth

Depending on the base frequency f_0 the width of the frequency group/critical band can be seen
About 100Hz up to 500Hz,
for higher frequencies (20kHz) up to 4kHz bandwidth

Frequency Groups

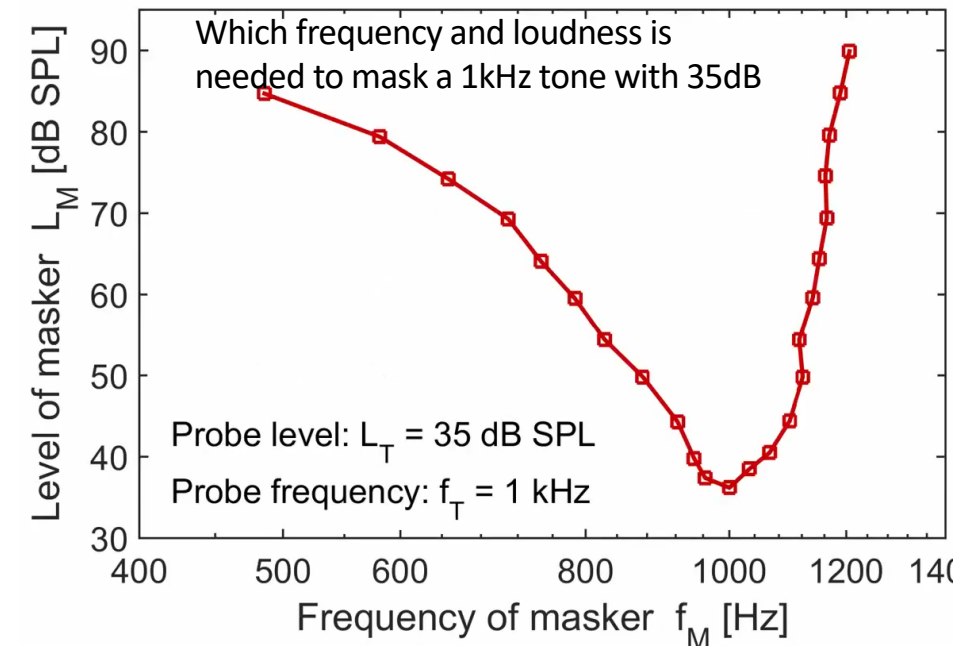
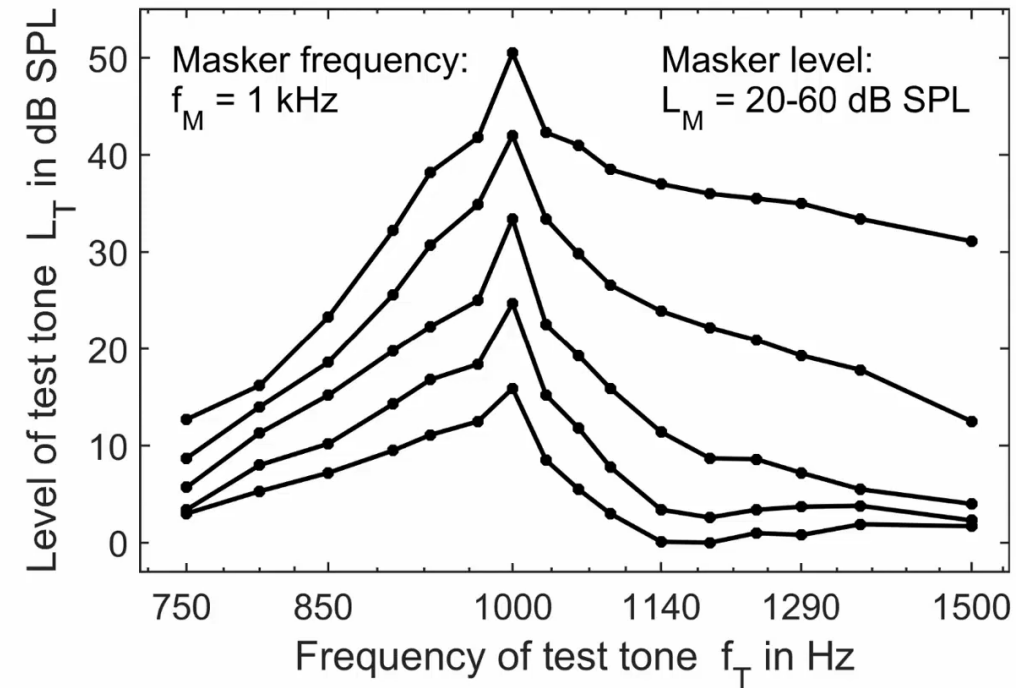
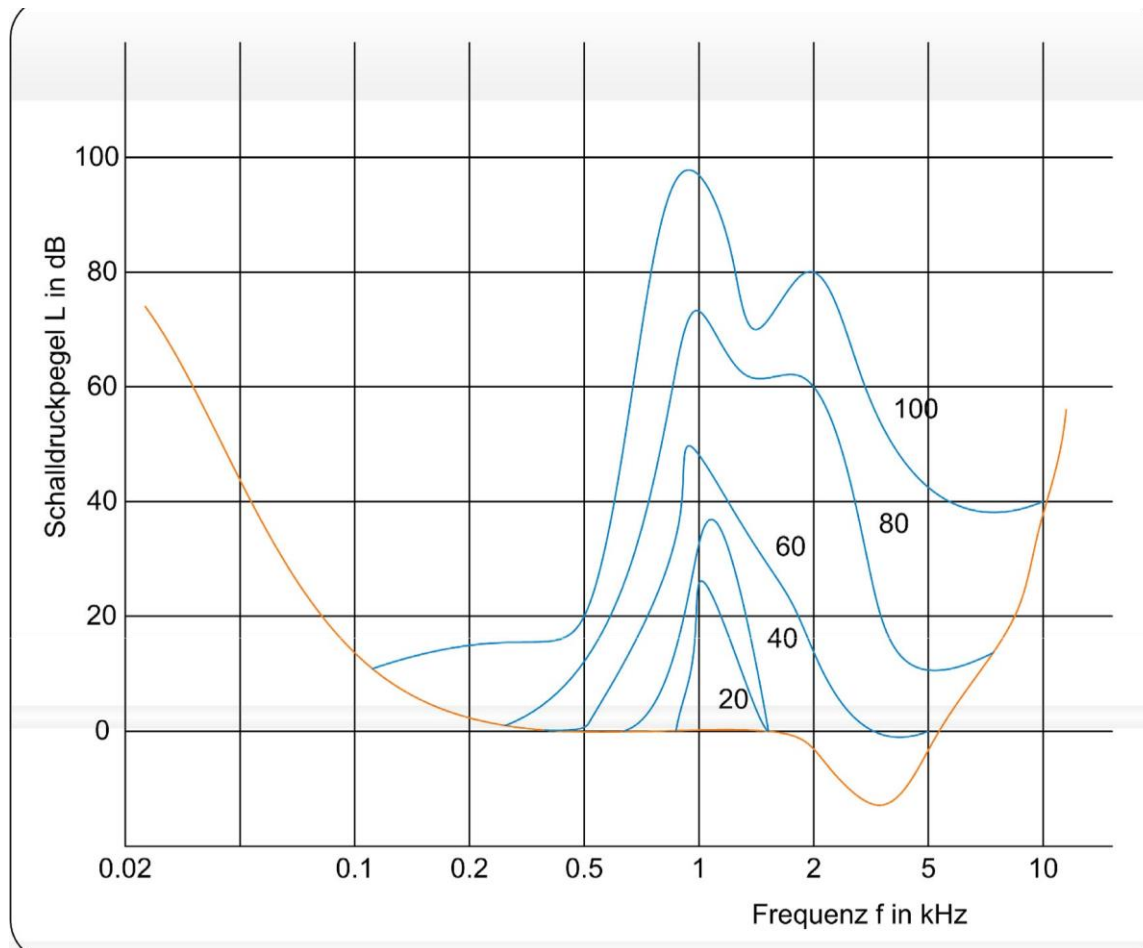
- Psychoacoustic effect of hearing
- Different perception of sounds within frequency bands:
 - Loudness
 - Frequencies of simultaneously sounding tones must have certain minimum distance to be perceived as tones of different pitch
 - Enable masking
- Frequency group comprises a frequency range within which a second tone will interfere with the perception of the first tone
 - refers only to simultaneously played tones (with sequentially played tones, frequency resolution of the auditory system is much higher).

Masking Effect

- Two tones play simultaneously
 - with different volume and
 - "similar" frequency
- Masking effect - ear hears only the loud sound
 - Quiet sound is "masked"
 - the loud sound reduces sensitivity of hearing in the surrounding frequency range
 - More quiet sound is below new hearing threshold curve
- Masking threshold is minimum volume that the more quiet sound would have to have in order not to be masked

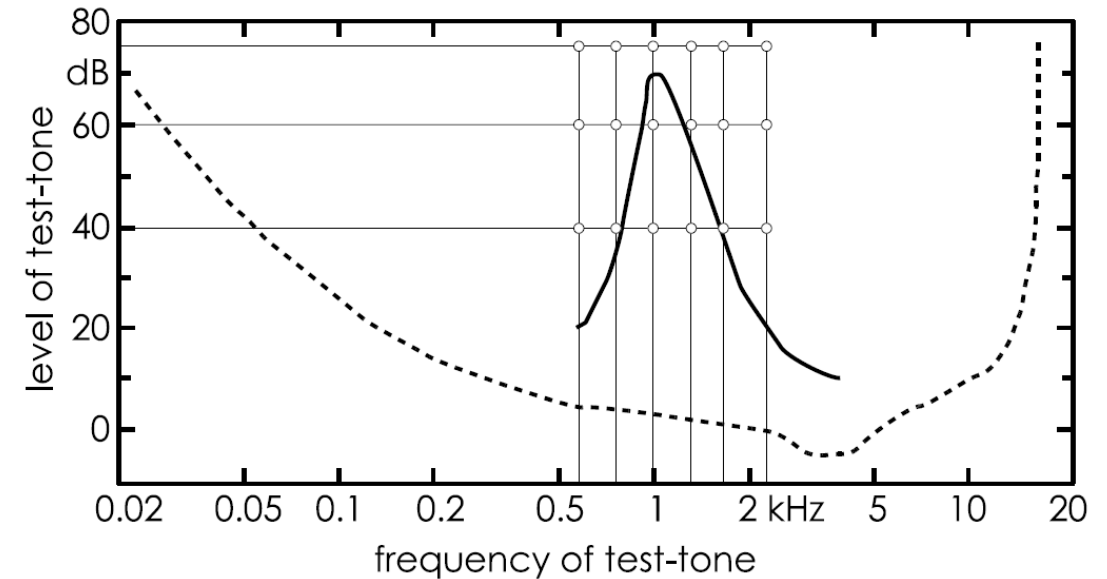
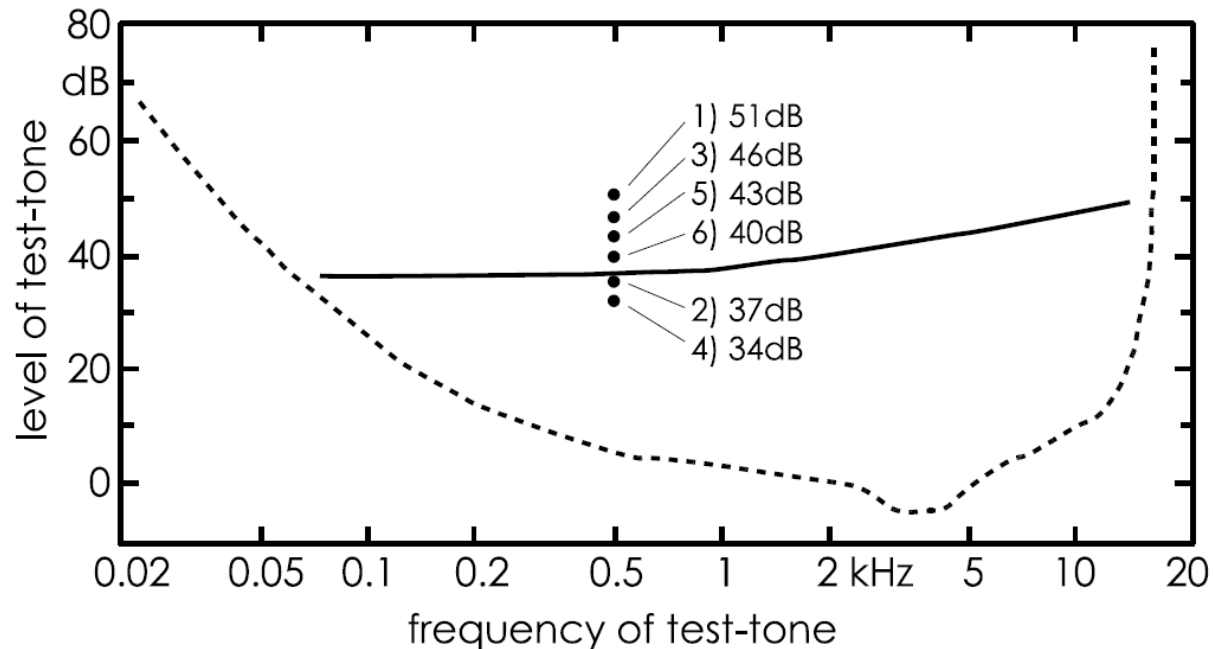
Masking Effect

Hearing threshold curve at 1kHz maskers
(different sound intensities)





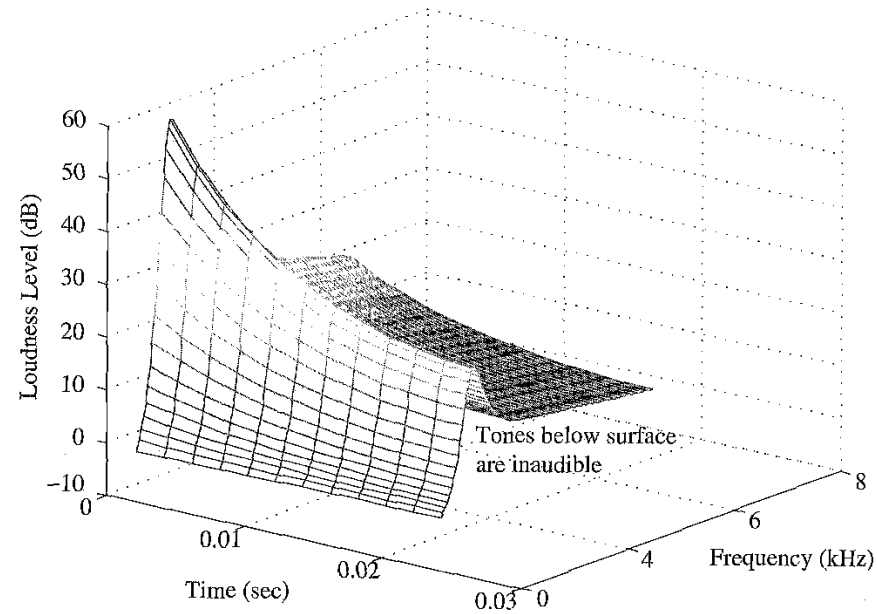
Masking - Examples



A pure tone is masked by white noise. The tone has a frequency of 50 Hz, the white noise has an overall level of 63 dB which corresponds to a density level of about 20 dB. You will hear six presentations, each with triplets of pure tones, masked by white noise. The level of the tone triplets is (in this order) 51 dB, 37 dB, 46 dB, 34 dB, 43 dB, 40 dB. Except for the second (37 dB) and fourth triplet (34 dB), the tones are above masked threshold and should be audible.

The masked threshold of pure tones masked by critical-band wide noise (1 kHz, 70 dB) is illustrated. You will hear three series of tone triplets: the first series is played at a level of 75 dB, the second at a level of 60 dB, the third at a level of 40 dB. Each series consists of six tone triplets with the frequencies 600 Hz, 800 Hz, 1000 Hz, 1300 Hz, 1700 Hz, and 2300 Hz. In the second series the third tone triplet at 1000 Hz is masked by the narrow-band noise, and in the third series the third and fourth triplet at 1000 Hz and 1300 Hz (for some persons also the fifth triplet at 1700 Hz) are masked.

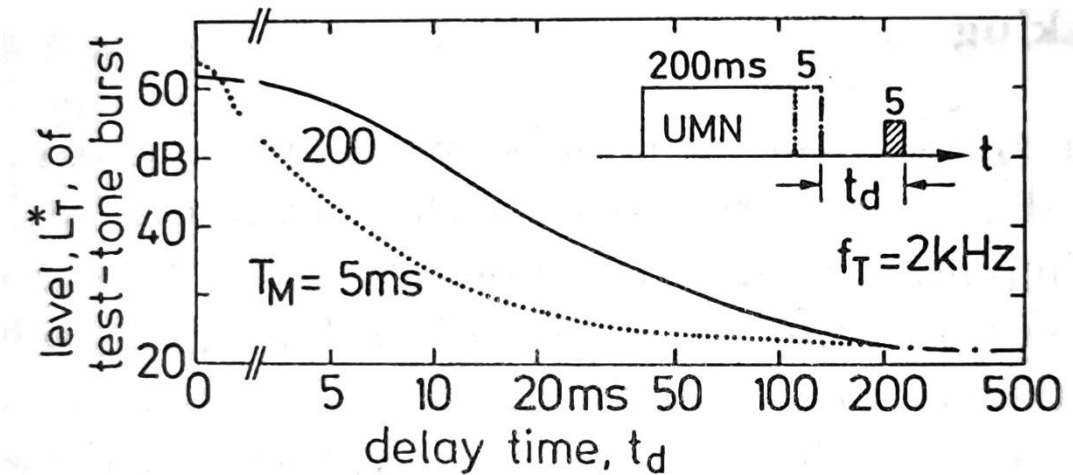
Masking



- Masking depends on volume, frequency, time interval and duration of the masking sound ("masker").

Temporal Masking

- Pre-masking (up to 2-20 msec)
 - Tone before a louder tone is masked
- Post-masking (50-200 msec)
 - Tone after a louder tone is masked



The level of the uniform masking noise is 60dB; duration of the 2-kHz test tone is 5ms (dotted line). Test tone of 200ms – solid line.

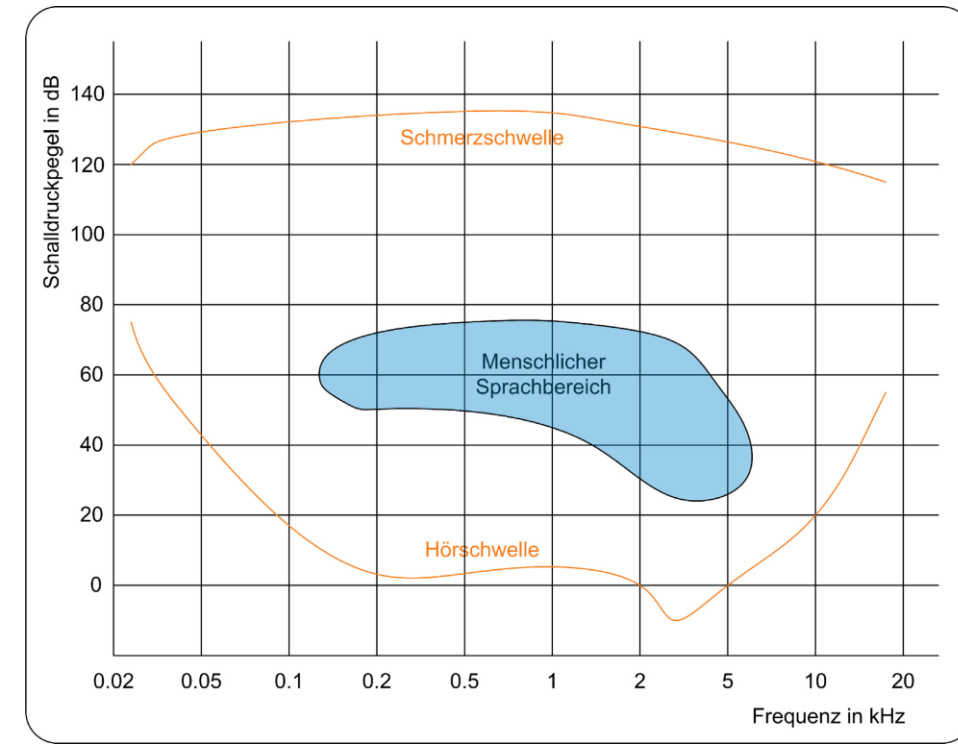
Note that delay time in this case is time between the end of the masker and the end of the test tone.

Postmasking depends on masker duration:

Levels of just-audible test-tone bursts are indicated as a function of their delay time after the offset of maskers of 200-ms and 5-ms duration.

Listening Range – MPEG Audio Compression

- Pain threshold and speech range are not relevant for MPEG audio coding
- Hearing threshold - essential for MPEG audio coding
 - Sound must have certain minimum sound level to be perceived
- Minimum sound level is frequency-dependent
- Hearing threshold curve: minimum sound level/frequency function
 - Orange curve is resting hearing threshold (“Ruhehörschwelle”)



Masking and MPEG Audio

- General audio signal consists of many frequency components
 - each frequency component influences the hearing threshold curve
 - Audio signal varies with time
- Hearing threshold different at each point in time
- **Psychoacoustic model**
 - Calculates current hearing threshold curve in each case
- **Variable quantization**
 - Frequency components quantized so that quantization noise just below current hearing threshold curve
 - Where masking is used, coarse quantization, small word lengths

Psychoacoustic Effects - Demos

- Further **Examples in PDF** (TUWEL):
“*Psychoacoustic demonstrations*” from the book
 - Psychoacoustics: Facts and Models,
by Hugo Fastl and Eberhard Zwicker, 3rd ed., 2007.
- Patti Software (Psycho-Acoustic Toolbox at TU Ilmenau)
<https://sourceforge.net/projects/patti/>
- Shepard Skala
<https://mynoise.net/NoiseMachines/shepardAudioIllusionToneGenerator.php>

Cochlea Implant

- Short German description and hearing sample
<https://www.youtube.com/watch?v=iFY7cmDIWqE>
- <https://www.youtube.com/watch?v=8uthhsMncDY>

Digital Audio

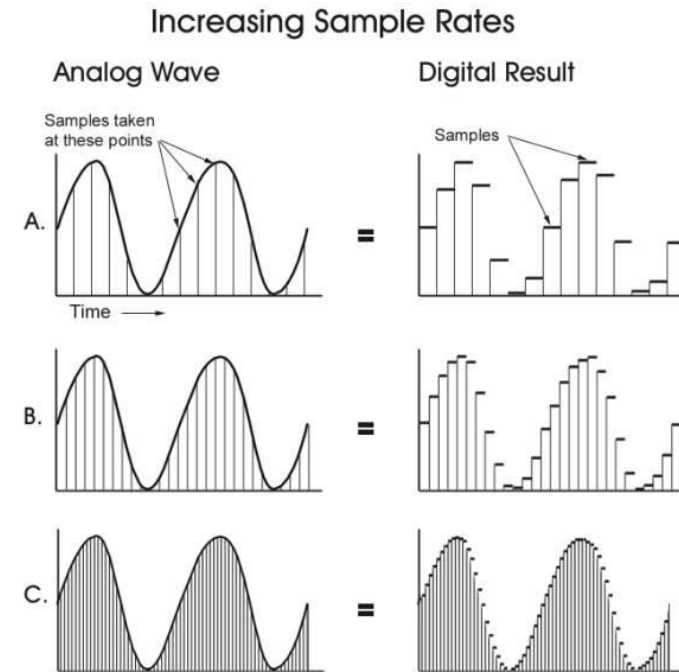
Introduction Digital Audio

Microphone converts sound wave into electrical signal: voltage curve analogous to the pressure fluctuation curve at the diaphragm of the microphone.

- Analog audio technology:
Analog signal stored on magnetic tape, vinyl record...
Editing: mechanical cutting, dubbing
- Digital audio technology:
Analog signal is converted into binary representation by A/D converter, stored, edited on computer; is output again as analog signal at loudspeaker by means of D/A converter.

Digitization in General: Sampling & Quantization

- Microphones produce analog signals (continuous-valued voltages)
- To get audio into a computer, it must be converted into a stream of numbers;
discrete sampling (both time and voltage)
- **Sampling**: divide the horizontal axis (**the time dimension**) into discrete pieces; uniform sampling is ubiquitous
- **Quantization**: divide the vertical axis (**signal amplitude**) into pieces; sometimes, a non-linear function is applied

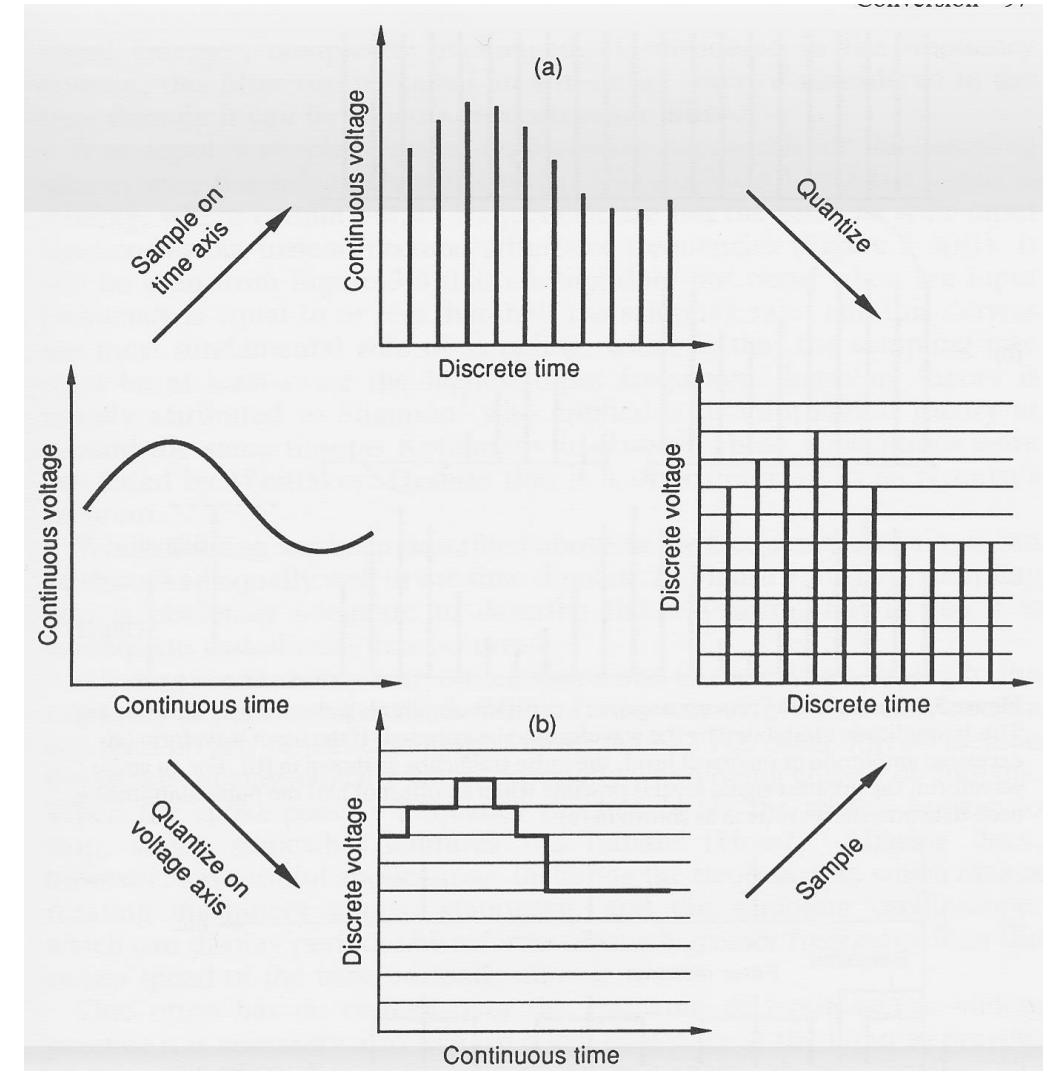


Digitization Basics

- Input: time and voltage continuous waveform
- Sampling- Quantizing: are independent of each other temporal order does not matter
- Result: time and voltage discrete format

(a) Input signal is sampled, samples are quantized (common with audio)

(b) Input signal is quantized, then sampled (usual for video)



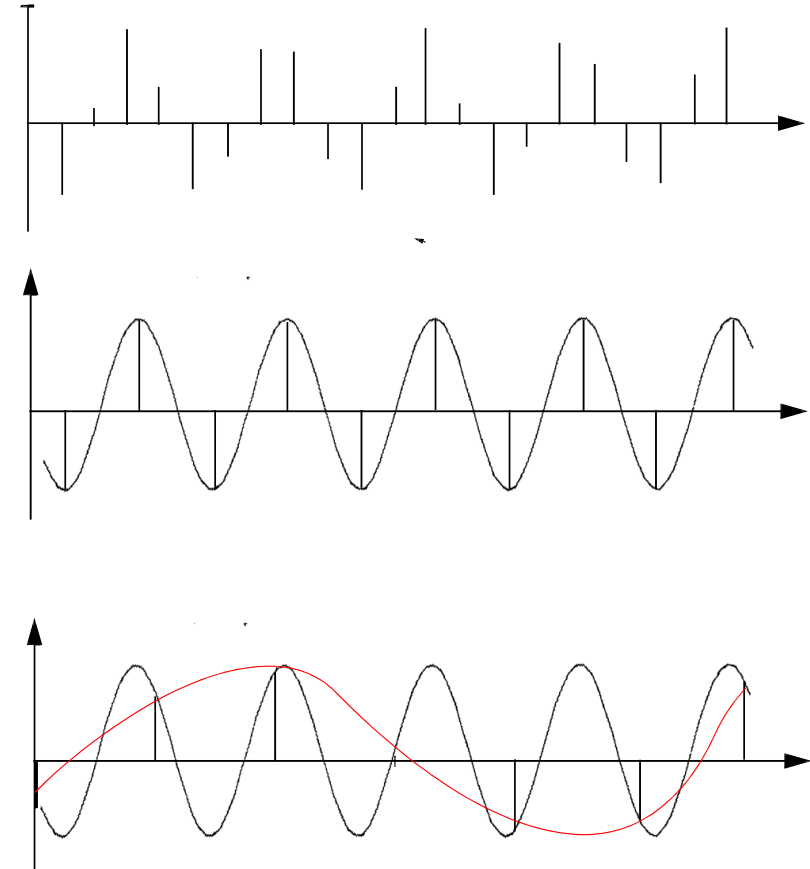
Digitization in General

- Questions for producing digital audio (Analog-to-Digital Conversion):
 - how often do you need to sample the signal?
 - how good is the signal?
 - how is audio data formatted?
- Suppose we are sampling a sine wave.
 - How often do we need to sample it to figure out its frequency?
 - If we sample at 1 time per cycle, we can think it's a constant
 - If we sample at 1.5 times per cycle, we can think it's a lower frequency sine wave --> alias effects

Alias component

Aliasing is an effect that causes different signals to become indistinguishable (or *aliases* of one another) when sampled.

Often refers to the distortion or artifact that results when a signal reconstructed from samples is different from the original continuous signal.



Sampling: Nyquist-Shannon Theorem

- Procedure: a sample is taken from the analog input signal at periodic time intervals.
- How often must an analog signal be sampled?

Nyquist-Shannon Theorem:

A sampled signal can be reconstructed without loss of information, if and only if

$$f_s \geq 2 \times f_{max}$$

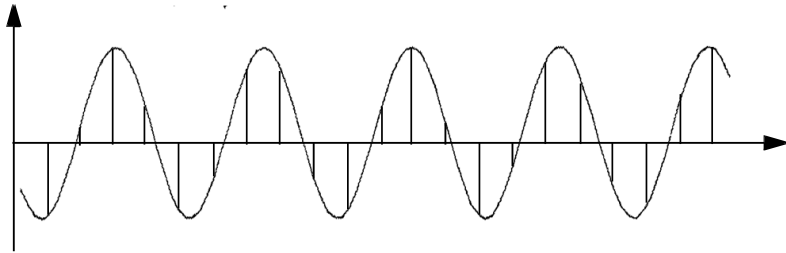
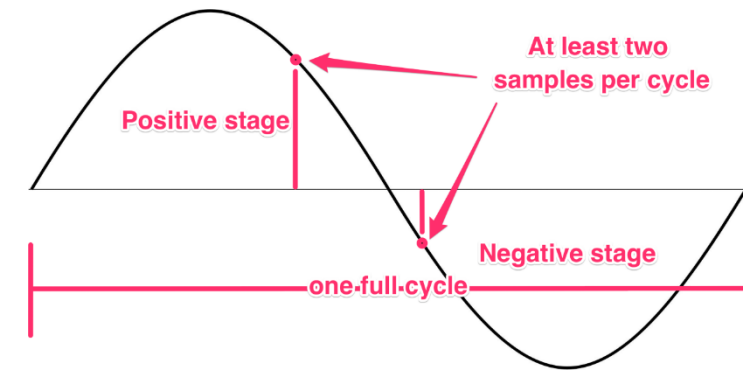
f_s Sampling frequency

f_{max} ...the highest frequency occurring in the signal

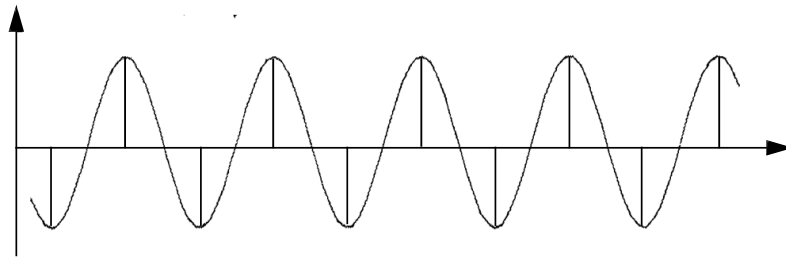
Nyquist Rate, Nyquist Frequency

Alias Component

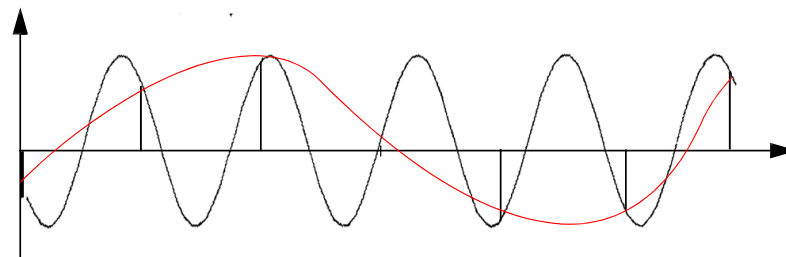
To determine the wavelength and therefore frequency of a sound wave we need at least 2 samples per cycle.



- $f_s > 2 \times f_{max}$
Reconstructed signal identical with original

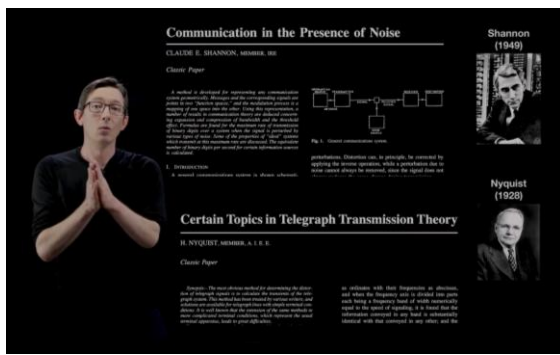


- $f_s = 2 \times f_{max}$
Reconstructed signal same f as original



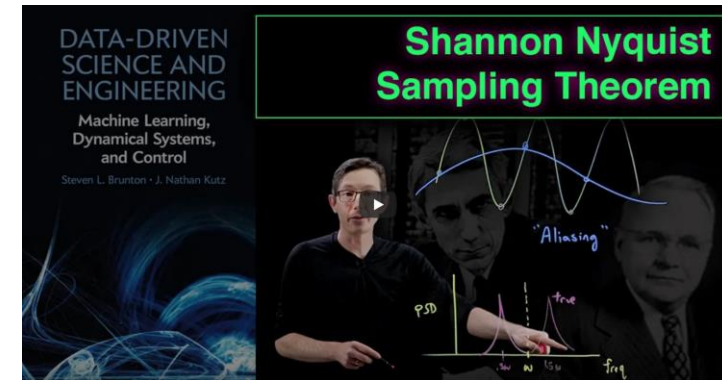
- $f_s < 2 \times f_{max}$
Interference frequency
"alias component"

Ex. : $f_s = 48\text{kHz}$, $f_{id} = 30\text{kHz}$, $f_{alias} = 18\text{kHz}$

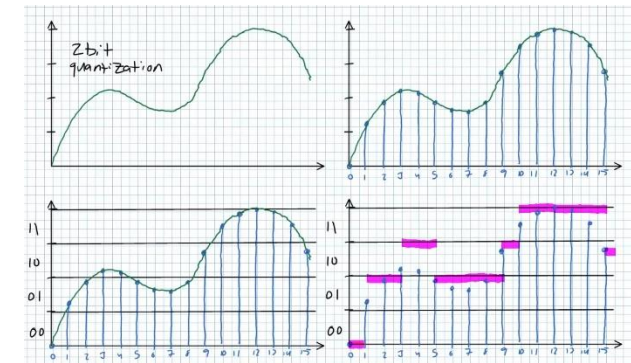


Optional Material

Shannon Nyquist Sampling Theorem
by Steve Brunton, History, Overview & Outlook
<https://www.youtube.com/watch?v=FcXZ28BX-xE>



Analog to Digital Conversion Basics
(2-4 bit quantization)
by Jim Pytel
https://www.youtube.com/watch?v=sRq_gM8r6pl



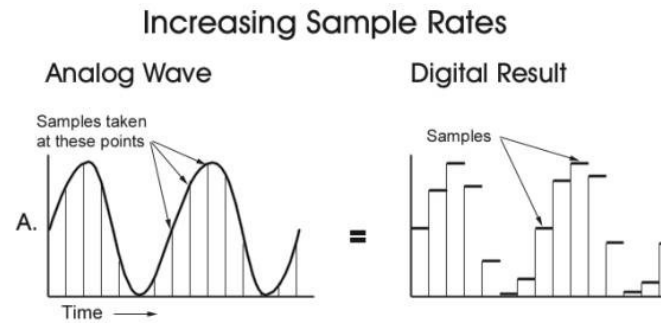
Optional (starting from middle)
Digital Signal Processing Basics and Nyquist Sampling Theorem
<https://www.youtube.com/watch?v=WgJMjDh0nLU>

Examples

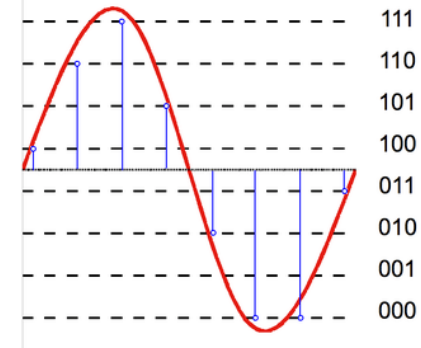
Sampling frequency and max. frequency

Format	Sampling frequency	Frequency range
Phone	8 kHz	200-3400 Hz
Audio CD	44.1 kHz	20-20000 Hz
Prof. Audio	48/96/192 kHz	20-20000 Hz
FM/Sat- Radio	22/32 kHz	20-15000 Hz

Quantization



3-bit resolution
($2^3 = 8$ levels)



- Sampled voltage values are assigned discrete numerical values:
Total voltage range divided into quantization intervals Q
- ➔ Continuous value assigned to **nearest numerical value**.
- Quality: Q should be as small as possible – is a result of the length of the data word size:

V_{pp} = peak to peak range (max range of signal) or $2 * \text{abs}(V_{max})$

8 bit (256 intervals):
$$Q = \frac{V_{pp}}{2^8} = (4 * 10^{-3}) * V_{pp} = 0.004 * V_{pp}$$

16 bit (65536 intervals):
$$Q = \frac{V_{pp}}{2^{16}} = (15 * 10^{-6}) * V_{pp} = 0.000015 * V_{pp}$$

24 bit (16 777 216 intervals) is also used

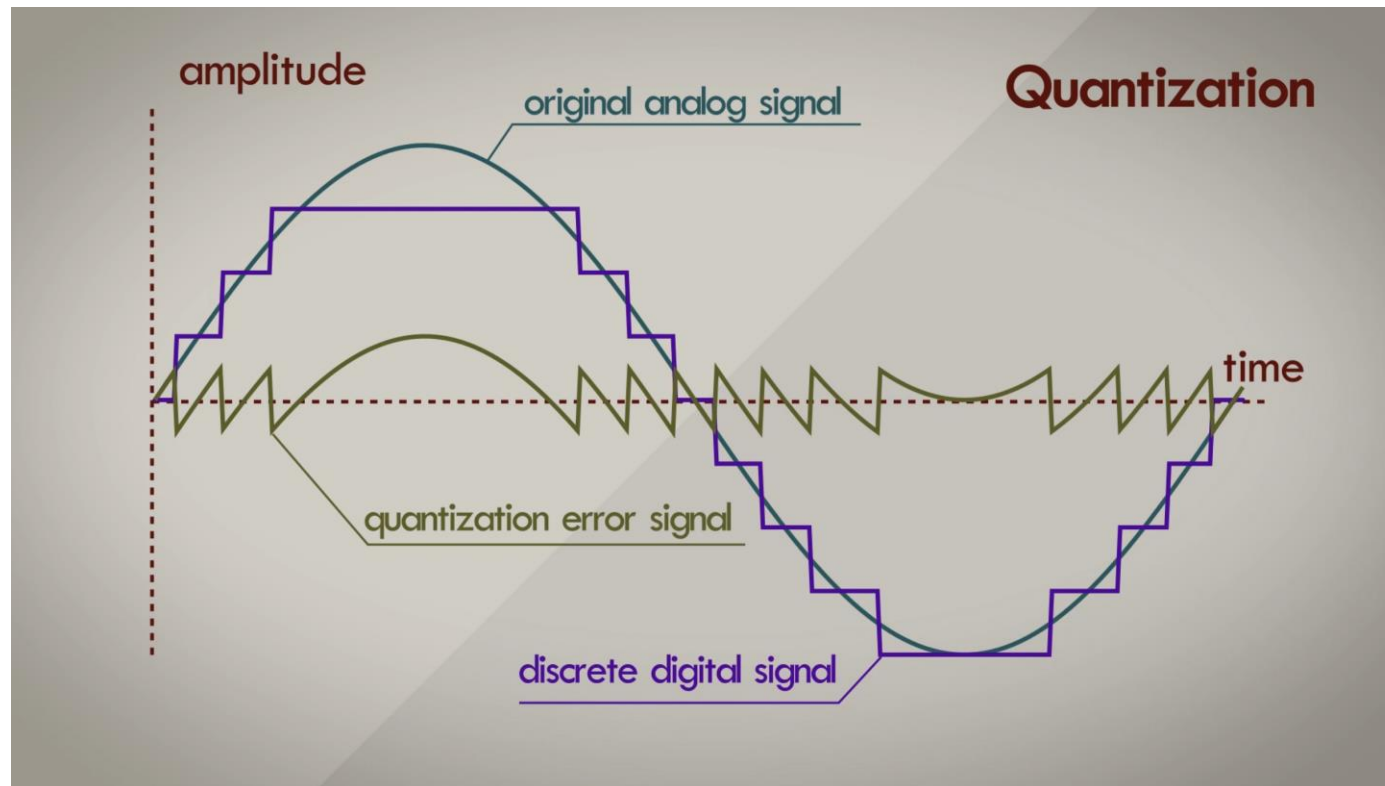
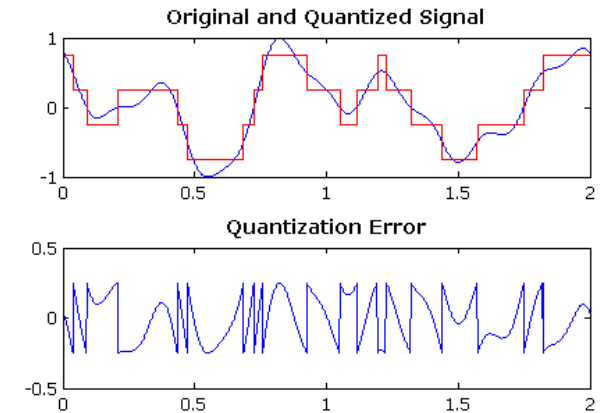
Some Examples

- 44,000 Hz 16 bit
110 KB
- 44,000 Hz 8 bit
90 KB
- 22,000 Hz 16 bit
61 KB
- 22,000 Hz 8 bit
45 KB
- 8,000 Hz 16 bit
16 KB
- 8,000 Hz 8 bit
13 KB



Quantization Error

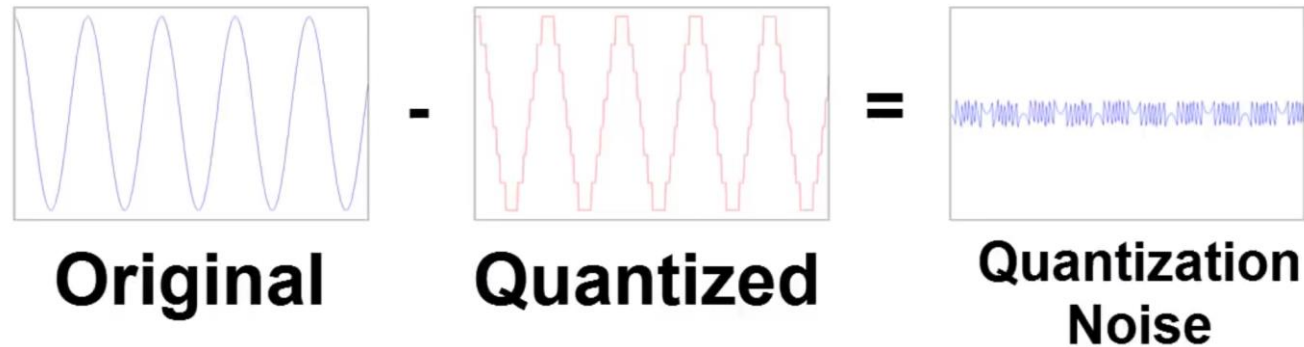
- Numerical value deviates from the sampled value by $\leq 0.5 Q$.
Error is audible as "quantization noise" during reconstruction.



<https://www.youtube.com/watch?v=1KBLguIXL30>

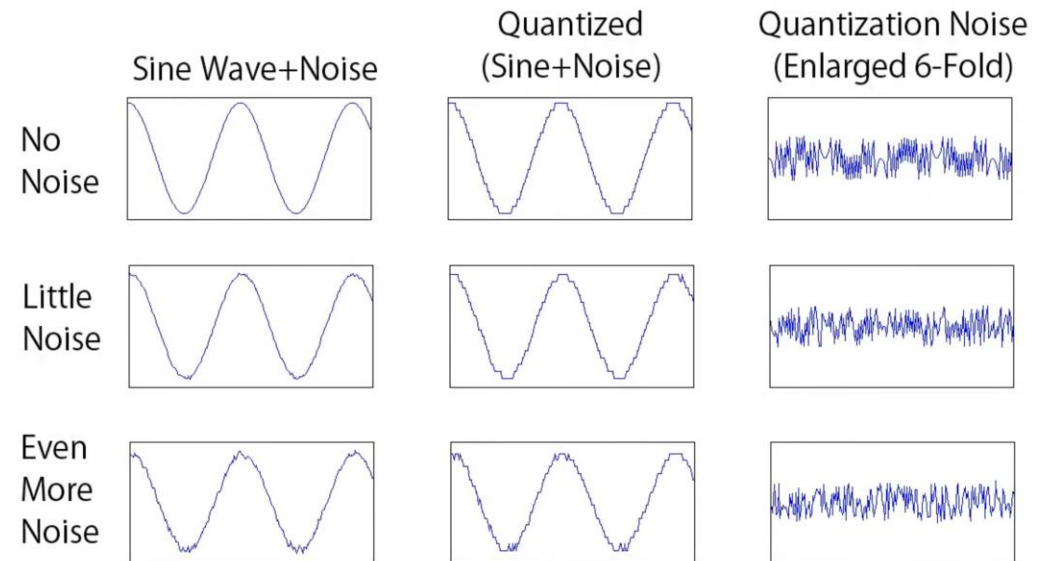
Quantization Noise & Dithering

- Quantization Noise



<https://www.youtube.com/watch?v=U2mwXiJqAgA>

- Dithering: intentionally adding noise during quantization to preserve low level information and prevent distortion



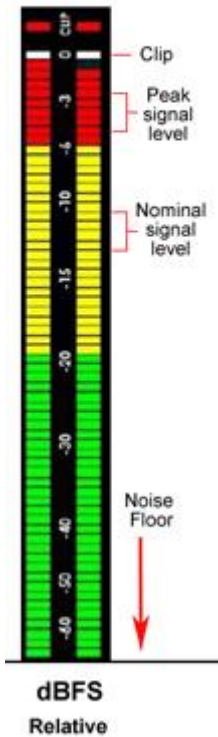
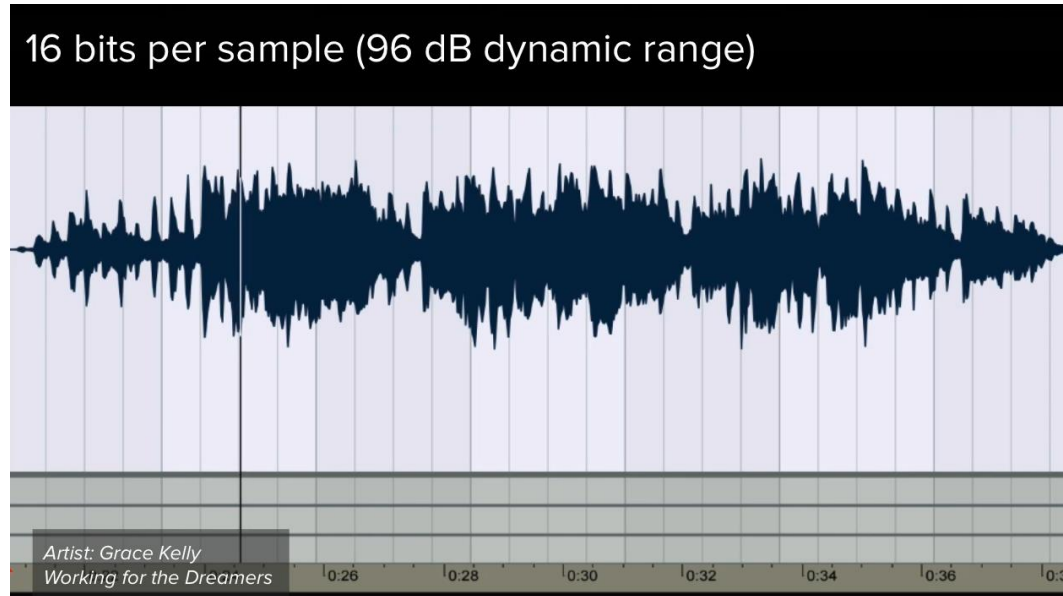
<https://www.youtube.com/watch?v=tb6X3WZ3-Uc>

Signal to Noise Ratio / Dynamic range

- **SNR**..Signal to noise ratio [dB] – in General:
 - Ratio of power of a signal (meaningful input) to power of background noise (unwanted input)
 - $SNR = \frac{P_{Signal}}{P_{Noise}}$
- In Audio:
 - Quantization noise generates sound pressure level L_{noise}
 - Maximum usable signal Sound pressure level L_{max} (loudest possible signal)
 - $SNR = \log \left(\frac{P_{Signal}}{P_{Noise}} \right) = L_{max} - L_{noise}$
- In audio **Dynamic Range** is SNR:

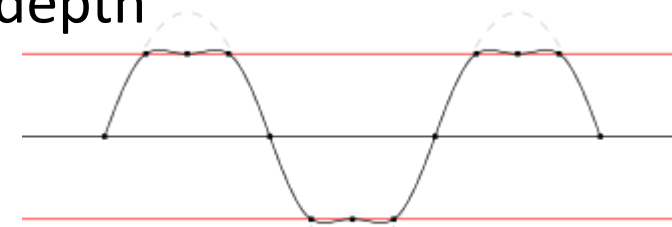
Resolution	8 bit	12 bit	16 bit	24 bit
Dynamic Range	49.8 dB	73.7 dB	96.3 dB	144.5 dB

Bit Depth – Dynamic Range



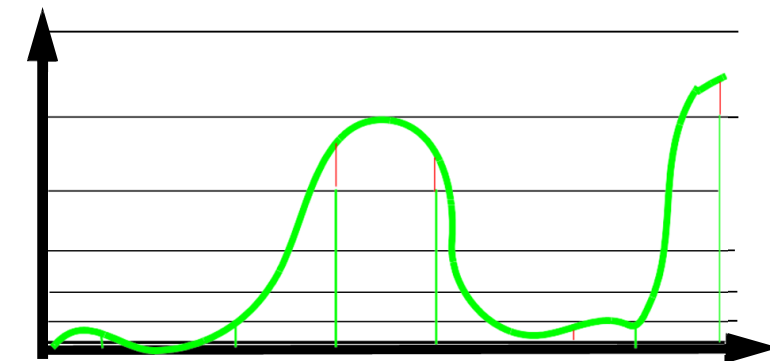
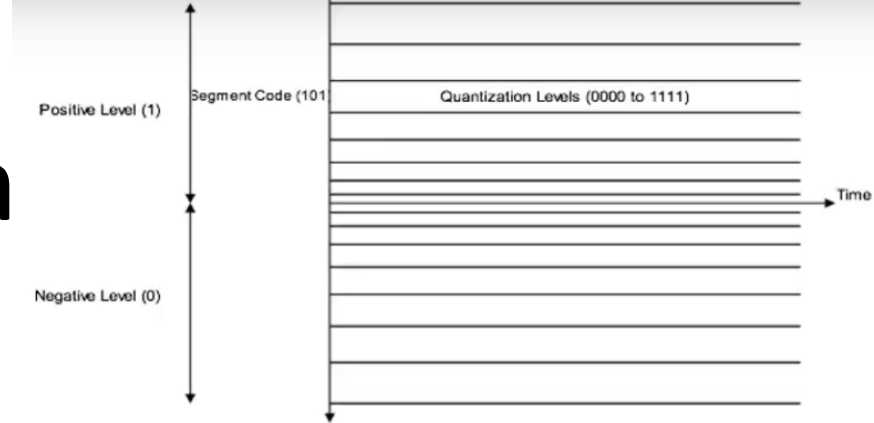
<https://www.youtube.com/watch?v=ubCMI3Jq6e4&list=PL4dISychPQEtnZkzPUfstKVYsG0e5gwbt&index=2>

- Remark: **dBFS = Decibel relative to Full Scale**, used in audio production software
= maximum peak level for digital signal according to bit depth
- Clipping occurs above 0 dBFS

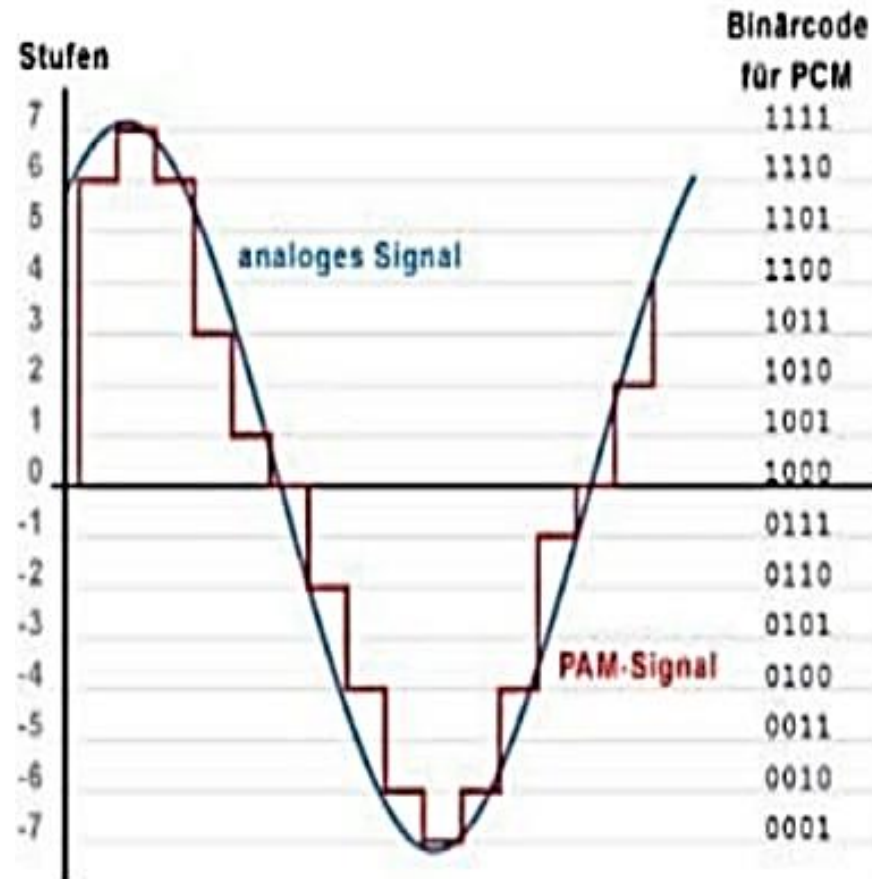


Linear; Non-linear quantization

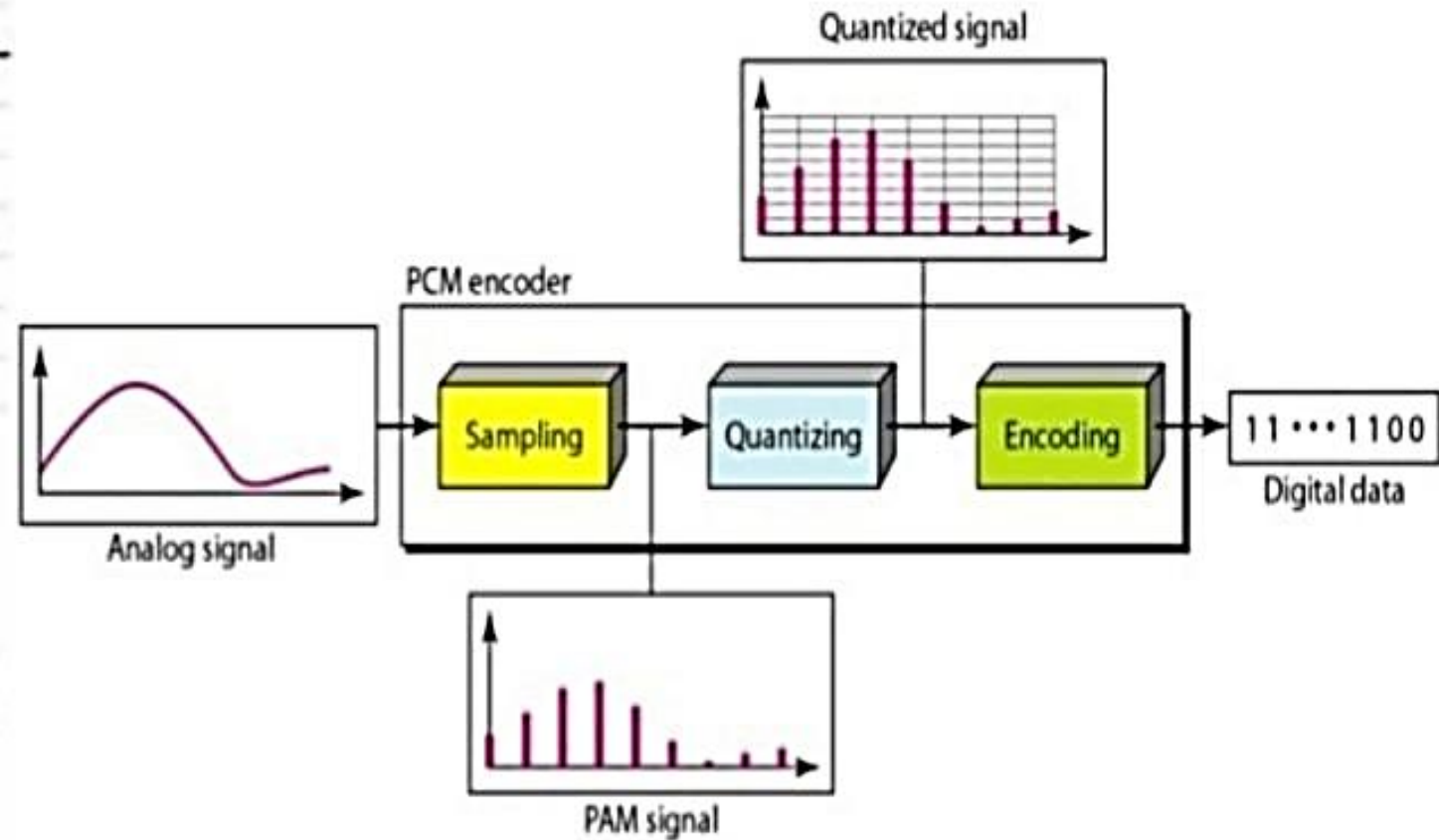
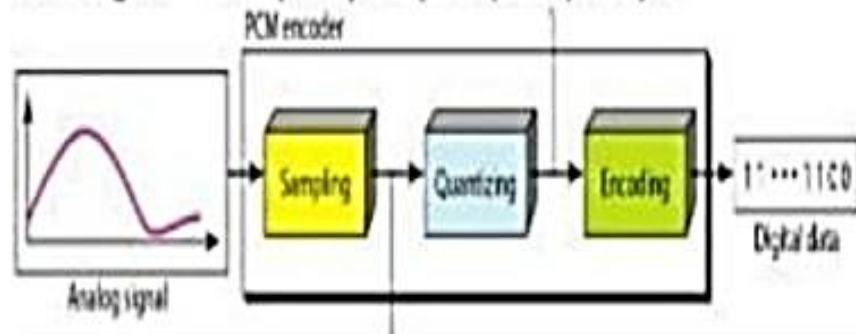
- Linear quantization: Q constant; common method in audio engineering
- Nonlinear quantization: Q of different size
small values $\rightarrow$ small quantization error
large values $\rightarrow$ large quantization error
- Noise masked if high desired signal $\rightarrow$ passable quality even with small resolution
- Usage for systems with low transmission bandwidth



PCM Pulse Code Modulation

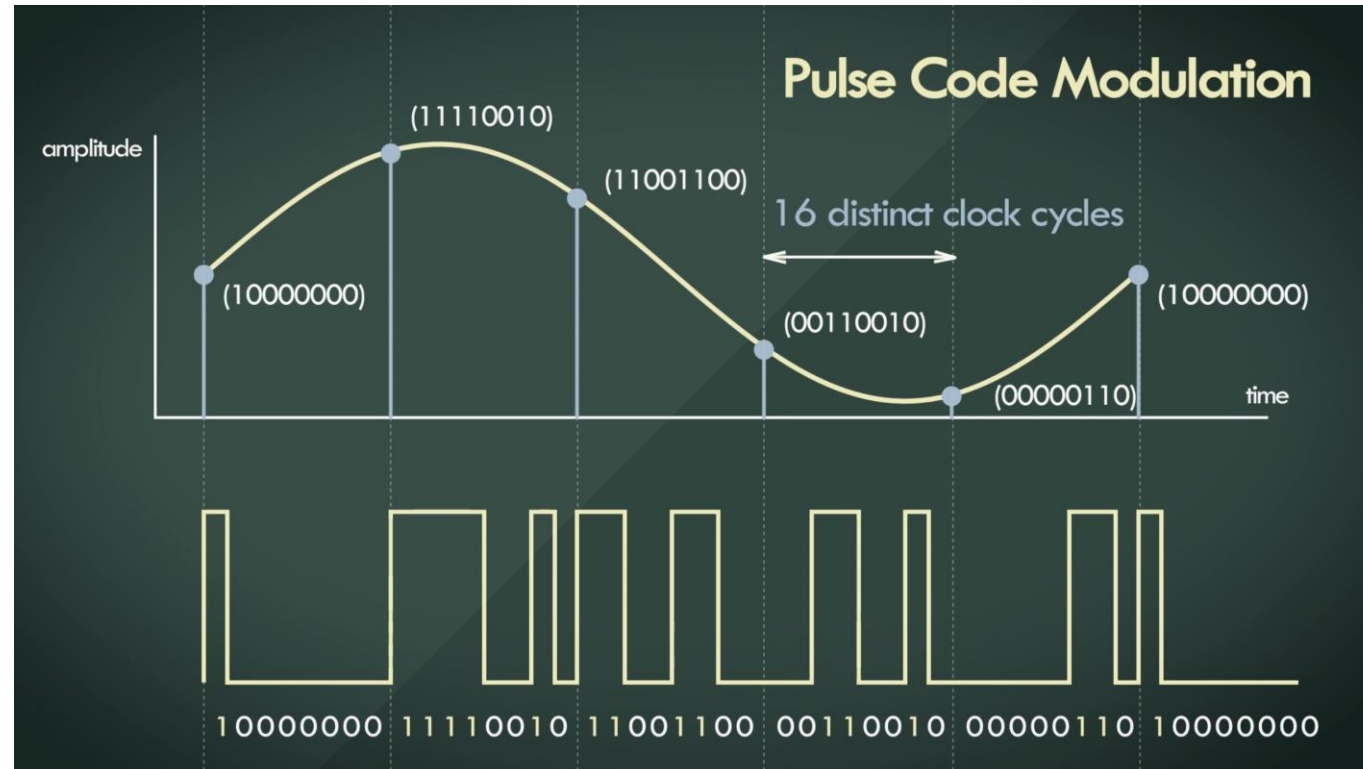


PCM-Signal 1110|1111|1110|1011|1001|1000|...



Pulse Code Modulation

- Sampling and Quantization give discrete values at discrete intervals
- In Pulse Code Modulation a discrete value is converted into a series of binary digits



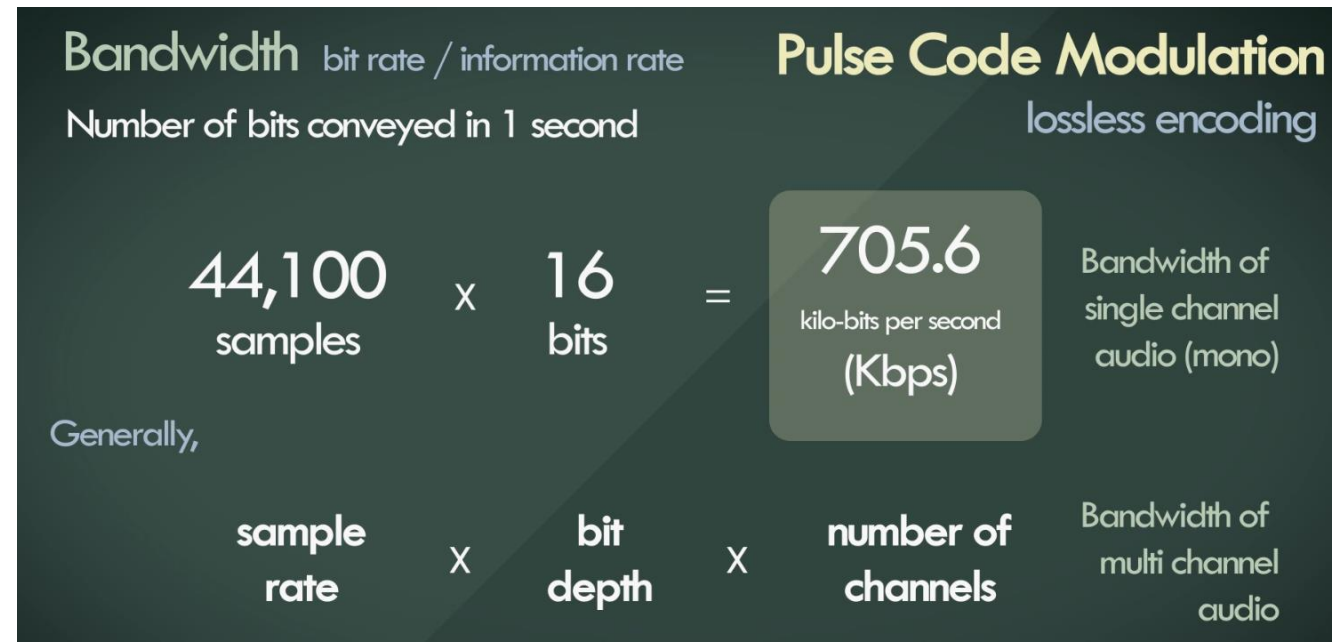
Pulse Code Modulation – Lossless Encoding

- Procedure: Binary values are transmitted serially in the form of modulated voltage pulses via a single line
- Disadvantage: **High data rate**

Sampling frequency [Hz] * bit depth [bit] = data rate [bit/s].

Ex. : CD-Audio(mono):

44,1kHz; 16 bit -> 705600 bit/s



Characteristics of Digital Audio

- sampling frequency (rate)
- sample size / quantization
- sample representation (negative values)
- number of channels (tracks)
- **Interleaving** (e.g. 2 channels: LRLRLRLRLR)
- coding/compression

Example Audio Formats

	CD audio	CD-I (Level B)	A-law ^a -law
sampling rate (kHz)	44.1	37.8	8
sample size (bits)	16	4	8
quantization	uniform	uniform	log
no. of channels	2	1-8	1
data rate per channel (bit/sec)	705	174	64
encoding	PCM	ADPCM	PCM
quality	very high	"Mid Fi", "FM"	"telephone"

a. A-law is used within European telephone systems, while -law is North American.

Adaptive differential pulse-code modulation (ADPCM) (1973) is a variant of **differential** pulse-code modulation (DPCM) that varies the size of the quantization step, to allow further reduction of the required data bandwidth. The adaptation to signal statistics in ADPCM consists simply of an **adaptive scale factor** before quantizing the difference in the DPCM encoder. Typical compression ratios are from 4:1 to 2:1.

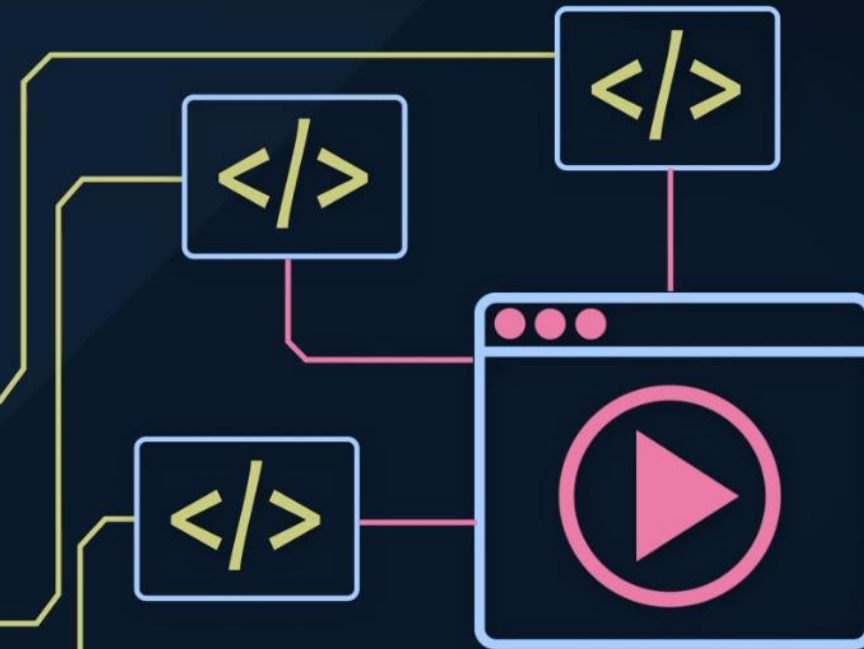
WAV File Format, FLAC & Codecs

Format chunk

compression type

code	description
0	unknown
1	PCM (uncompressed)
2	Microsoft ADPCM
6	ITU G.711 α -law
7	ITU G.711 μ -law
20	ITU G.723 ADPCM
49	GSM 6.10
80	MPEG
65,536	experimental

coder - decoder



Detailed Further Material

- Course on “**Digital Audio Fundamentals**” (12 parts)

https://www.youtube.com/playlist?list=PLbqhA-NKGP6B6V_AiS-jbvSzdd7nbwwCw

by Akash Murthy



Digital Audio Fundamentals

12 Videos • 1.185 Aufrufe • Zuletzt am 29.01.2021 aktualisiert

This video series explains the fundamentals of digital audio, how audio signals are expressed in the digital domain, how they're converted and transformed and the advantages of working with digital signals. We'll cover concepts of sampling theorem, sample rates, aliasing, bit depth, quantization, dither, pulse code modulation, sigma delta modulation, noise shaping and all that good stuff.

Akash Murthy **ABONNIEREN**

1	SIGNAL PATHS	1. Signal Paths - Digital Audio Fundamentals	Akash Murthy
2	SAMPLING THEOREM	2. Sampling Theorem - Digital Audio Fundamentals	Akash Murthy
3	COMMON AUDIO SAMPLE RATES	3. Common Audio Sample Rates - Digital Audio Fundamentals	Akash Murthy
4	ALIASING	4. Understanding Aliasing - Digital Audio Fundamentals	Akash Murthy
5	QUANTIZATION	5. Quantization - Digital Audio Fundamentals	Akash Murthy
6	BIT DEPTH	6. Bit Depth - Digital Audio Fundamentals	Akash Murthy
7	DITHERING EXPLAINED	7. Dithering Explained - Digital Audio Fundamentals	Akash Murthy
8	DITHERING TYPES & TECHNIQUES	8. Dither Types - Digital Audio Fundamentals	Akash Murthy
9	NOISE SHAPING	9. Noise Shaping - Digital Audio Fundamentals	Akash Murthy
10	PULSE CODE MODULATION	10. Pulse Code Modulation - Digital Audio Fundamentals	Akash Murthy
11	MULTIPLEXING & ERROR CORRECTION	11. Multiplexing and Error Correction - Digital Audio Fundamentals	Akash Murthy

MIDI = Musical Instruments Digital Interface

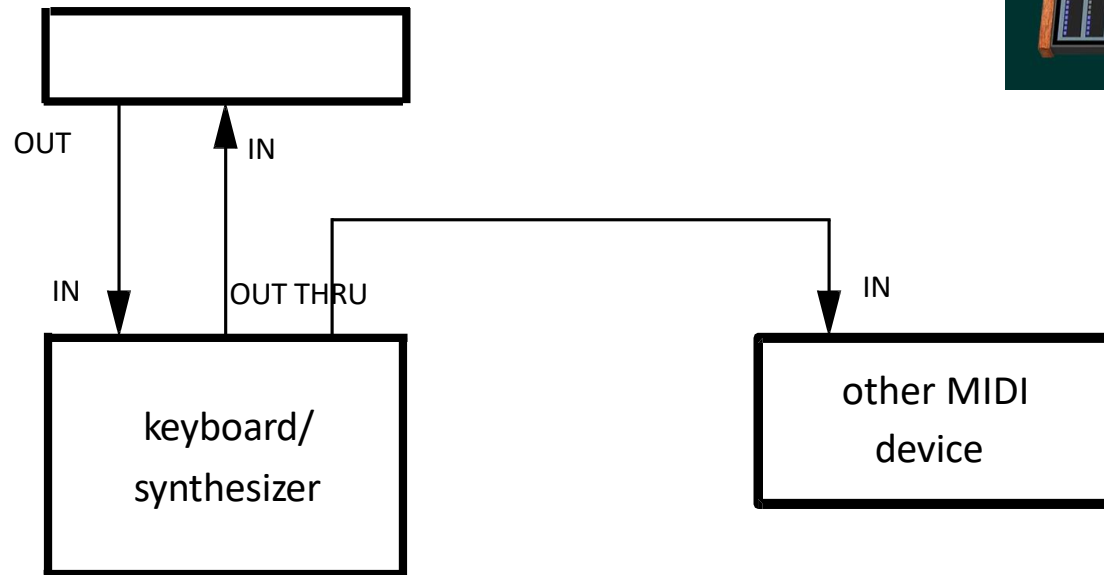
Terminology

- MIDI - a music description language in digital (binary) form
A protocol that enables computers, synthesizers, keyboards, and other musical devices to communicate with each other
- Digital audio workstation (DAW) - software used for recording, editing and producing audio files. It interfaces with MIDI devices (multiple MIDI INs and MIDI OUTs), can handle (edit/manipulate) MIDI data; uses plugins (e.g. based on VST standard) that provide virtual instruments and effects.
- Synthesizer - a sound generator (various pitch, loudness, tone color); usually has a microprocessor, keyboard, control panels, memory, etc.

MIDI = Musical Instruments Digital Interface

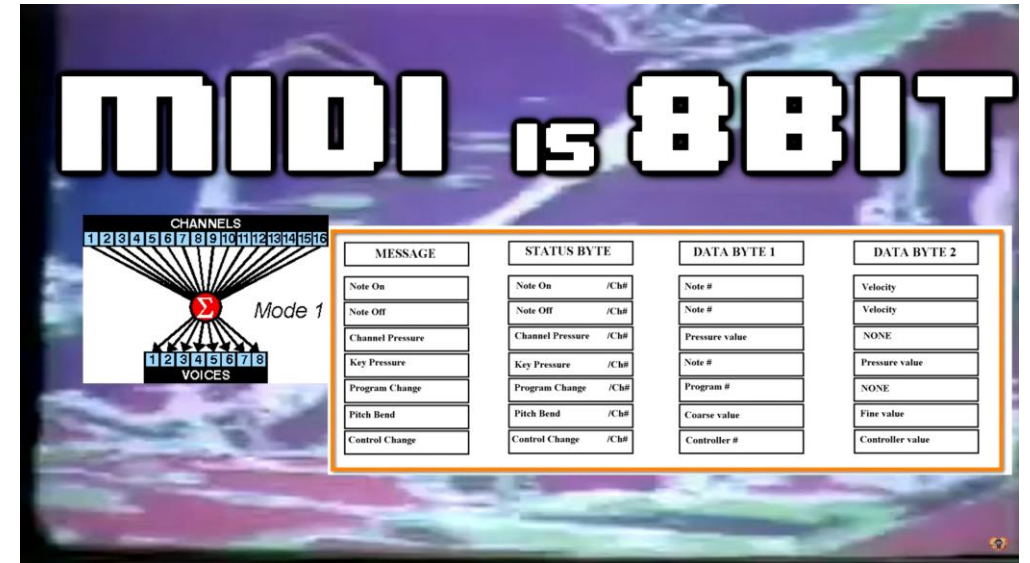
Example MIDI system configuration

Digital Audio Workstation (DAW)
– formerly a MIDI sequencer



MIDI – Brief History

First specification in 1983

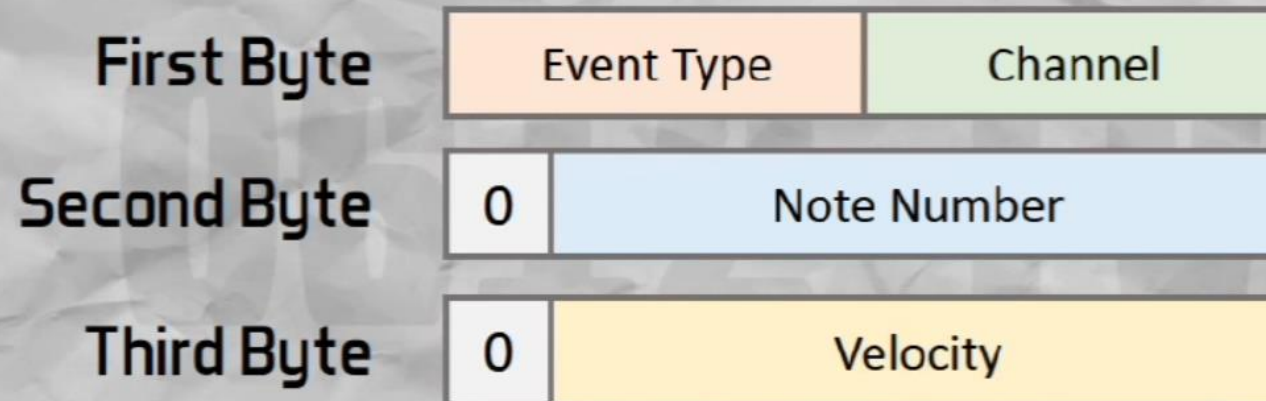


<https://www.youtube.com/watch?v=SUUxmJ84dni>

Basics of MIDI

Anatomy of a MIDI Message

Note On Event



<https://www.youtube.com/watch?v=5ZXuXqHdHks>

MIDI Example



MIDI Summary & Concepts

- MIDI is control data for electronic instruments and devices
- MIDI does not include audio signals
- Instruments create audio based on MIDI data
- MIDI messages are usually 1-3 bytes, each component 8-bit

Important MIDI Concepts:

- Timing clocks - MIDI sequencer time stamps messages, timebase measured in parts per quarter note, tempo in beats per minute
- MIDI synchronization - external or internal sync
- MIDI Time Code (MTC) - used to synchronize with film or video

Extended MIDI Specifications

Extended specifications for

- MIDI Machine Control (MMC)
- MIDI Show Control (MSC)
- MIDI Time Code
- General MIDI
- Downloadable Sounds
- Scalable Polyphony MIDI

Open Sound Control (OSC)

- A **protocol** for **networking** sound synthesizers, computers, and other multimedia devices for purposes such as musical performance or show control. OSC's advantages include interoperability, accuracy, flexibility and enhanced organization and documentation.
- First specified in 2002
- **Alternative to MIDI when higher resolution and a richer parameter space is desired**
- Main features, compared to MIDI, include:
 - Open-ended, dynamic, URI-style symbolic naming scheme
 - Symbolic and high-resolution numeric data
 - Pattern matching language to specify multiple recipients of a single message
 - High resolution time tags
 - "Bundles" of messages whose effects must occur simultaneously

